

HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

GRADUATION THESIS

Student's engagement detection in online classes

PHAN MINH HÒA

hoa.pm225495@sis.hust.edu.vn

Program: Data Science and Artificial Intelligence

Supervisor: PhD. Trịnh Văn Chiến

Department: Computer Science

School: School of Information and Communications Technology

HANOI, 06/2026

HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

GRADUATION THESIS

Student's engagement detection in online classes

PHAN MINH HÒA

hoa.pm225495@sis.hust.edu.vn

Program: Data Science and Artificial Intelligence

Supervisor: PhD. Trịnh Văn Chiến

Signature

Department: Computer Science

School: School of Information and Communications Technology

HANOI, 06/2026

ACKNOWLEDGMENT

I am grateful to my supervisor, **PhD. Trịnh Văn Chiến**, for the guidance, careful feedback, and patience that shaped this thesis. I thank the lecturers and staff of the School of Information and Communications Technology at Hanoi University of Science and Technology for the foundation they gave me, and the creators of the CMOSE [1] and DAiSEE [2] datasets, whose work made this study possible. Finally, I thank my family and friends for their constant support, and acknowledge my own persistence in seeing this work through.

ABSTRACT

Online and hybrid education has turned the webcam into the main link between student and teacher, yet a teacher who faces many video tiles at once can no longer tell when a learner has stopped paying attention. Automatic engagement recognition addresses this by assigning a short webcam clip an ordinal level from disengaged to engaged. Prior works either encode each frame with a facial-behaviour toolkit or learn directly from raw pixels, and they report accuracy on a single dataset. Each line of work carries its own limitation. The former flattens the heterogeneous facial measurements and passes them through a single encoder, while the latter learns appearance specific to one dataset and transfers poorly to new recording conditions. A further common practice is to report accuracy, which on such imbalanced data can look healthy while the model does no better than a random guess. To address these limitations, this thesis preserves the structure of the facial signal instead of flattening it, and replaces single-dataset accuracy with an evaluation built for imbalanced ordinal labels and for transfer. The proposed hybrid model splits the OpenFace descriptor into behavioural feature groups, encodes each group with its own temporal encoder, and optionally adds a global I3D motion stream, training every stream with a small auxiliary head. Every model is then selected and compared on an ordinal agreement measure that majority prediction cannot inflate, and its generalisation is measured directly across datasets. An ablation over the full space of encoder assignments shows that the hybrid family outperforms the strongest single-encoder baseline. A test-only private set of webcam clips recorded in the wild was collected and hand-labelled, and a cross-dataset study shows that engagement models transfer poorly and that accuracy conceals this collapse while the ordinal measure exposes it. Trained on the combined datasets, the proposed hybrid model generalises best to the unseen private set, and its margin over the baseline widens out of domain, where the only test that matters is deployment on unseen data.

Keywords: student engagement recognition, proposed hybrid model, feature-group temporal fusion, ordinal classification, cross-dataset generalisation.

Student

(Signature and full name)

TABLE OF CONTENTS

CHAPTER 1. INTRODUCTION.....	1
1.1 Problem Statement.....	1
1.2 Background and Problems of Research	1
1.3 Research Objectives and Conceptual Framework	2
1.4 Contributions	3
1.5 Organization of Thesis	4
CHAPTER 2. LITERATURE REVIEW	6
2.1 Scope of Research	6
2.2 Related Work	6
2.3 Background Knowledge	7
2.3.1 Feature Extraction.....	7
2.3.2 Temporal Architectures.....	8
2.3.3 Loss Functions for Imbalance and Ordinality	11
2.3.4 Evaluation Metrics	12
2.3.5 Statistical Measures for Result Analysis	14
CHAPTER 3. METHODOLOGY.....	16
3.1 Overview	16
3.2 Data and Preprocessing	17
3.2.1 Public datasets	17
3.2.2 Private dataset	18
3.2.3 Preprocessing	19
3.3 Baseline Models	19
3.4 Proposed Hybrid Model	21
3.4.1 Grouping the facial features by meaning.....	21

3.4.2 Proposed model architecture	22
CHAPTER 4. EXPERIMENTAL RESULTS	25
4.1 Experimental Setup.....	25
4.1.1 Hardware Specification	25
4.1.2 Hyper-parameter Specification	25
4.1.3 Evaluation Protocol.....	26
4.2 CMOSE In-Domain Result	27
4.3 Cross-Dataset Result.....	29
4.4 Private Dataset Result	31
CHAPTER 5. CONCLUSIONS	35
5.1 Summary	35
5.2 Suggestion for Future Work	37
REFERENCE	40
APPENDIX	42
APPENDIX A. BASELINE MODELS	42
A.1 Experimental Setup.....	42
A.2 Evaluation Protocol.....	43
A.2.1 Selecting the Primary Metric.....	43
A.2.2 Selecting the Development Dataset	44
A.2.3 Selecting the Development Loss	44
A.3 In-Domain Results	45
A.4 Prediction Agreement.....	46
A.5 Cross-Dataset Generalisation	48

LIST OF FIGURES

Figure 2.1	Residual TemporalBlock	9
Figure 2.2	LSTM cell at time step t	10
Figure 2.3	Transformer encoder layer	10
Figure 3.1	The baseline and the proposed hybrid model.	16
Figure 3.2	Representative frames from the private set, illustrating the range of camera angles found in unconstrained online-class recordings.	18
Figure 3.3	The five single-encoder baselines.	20
Figure 3.3	The five single-encoder baselines (continued).	21
Figure 3.4	Proposed hybrid model architecture (I3D stream enabled; the I3D-stream-disabled variant drops the I3D row).	22
Figure 3.5	The three interchangeable per-group encoders.	23
Figure 4.1	Design-space sweep of the proposed hybrid model across all 243 per-group encoder configurations, with the I3D stream enabled and disabled, on all six metrics (in-domain CMOSE). The dashed line is the QWK-selected baseline (OpenFace TCN, cross-entropy).	27
Figure 4.2	Cross-dataset generalisation of the proposed model on all six metrics; each cell is the single QWK-selected best configuration, reported on every metric, so the panels profile the one model deployed in that cell.	30
Figure 4.3	Cell-by-cell advantage of the proposed model: QWK-selected best proposed-model minus QWK-selected best baseline over the 3×3 train $\times$ test matrix, on all six metrics (Figure 4.2 minus the baseline matrix, Appendix A, Figure A.5); green is always the proposed-model-favouring direction.	31
Figure 4.4	Per-class F1 on the Private set for the combined-trained best baseline versus best proposed-model configuration.	32
Figure 4.5	In-domain QWK (trained and tested on CMOSE) versus transfer QWK (trained on Combined, tested on the held-out Private set); each configuration is one point pairing its own two runs. Spearman $\rho = 0.27$ for the 486 proposed-model configurations and 0.31 for the 15 baselines.	33

Figure A.1	Rank agreement (Kendall τ) between the six metrics over the fifteen baseline configurations, in-domain on CMOSE (MAE metrics sign-flipped so positive τ means agreement).	44
Figure A.2	QWK against the balanced secondary metrics (macro-accuracy and macro-MAE) for every baseline across the three losses, in-domain on CMOSE.	45
Figure A.3	The five baseline models under the three losses on all six metrics, in-domain on CMOSE.	46
Figure A.4	Pairwise prediction agreement (Cohen's κ) between all fifteen baseline configurations on the in-domain CMOSE test set, in model-major blocks.	47
Figure A.5	Cross-domain generalisation of the best baseline per cell, on all six metrics; each cell is the single QWK-selected best baseline, reported on every metric.	48

LIST OF TABLES

Table 3.1	Dataset statistics and class balance (%); the Private set is test-only.	17
Table 3.2	Grouping of the 709-dimensional OpenFace descriptor into five feature groups by meaning.	22
Table 4.1	Per-group marginal effect of encoder choice on mean QWK, in-domain (CMOSE) and pooled over the unseen-target cells.	28
Table 4.2	Top-5 configurations of the proposed hybrid model, in-domain (CMOSE), sorted by QWK.	29
Table 4.3	Paired effect of the I3D stream on QWK, by evaluation regime.	31
Table A.1	Winning baseline configuration by metric (in-domain CMOSE).	43

LIST OF ABBREVIATIONS

Abbreviation	Definition
AU	Facial Action Unit
BiLSTM	Bidirectional Long Short-Term Memory network
CDF	Cumulative Distribution Function
CE	Cross-Entropy loss
CMOSE	Class-level Multimodal Online Student Engagement dataset
DAiSEE	Dataset for Affective States in E-Environments
EMD	Earth Mover’s Distance; here the squared-EMD / CDF-MSE ordinal loss
GELU	Gaussian Error Linear Unit
I3D	Inflated 3D ConvNet (spatiotemporal video feature extractor)
LSTM	Long Short-Term Memory network
MAE	Mean Absolute Error
MLP	Multi-Layer Perceptron
QWK	Quadratic-Weighted Kappa
ReLU	Rectified Linear Unit
SMOTE	Synthetic Minority Over-sampling Technique
TCN	Temporal Convolutional Network

CHAPTER 1. INTRODUCTION

1.1 Problem Statement

Online and hybrid classes are now well integrated into public education, where lessons run through video calls and the webcam is the main connection between a student and a teacher, as well as the main evidence of whether students are following the lesson or not. In a physical classroom a teacher observes students' expressions, postures, and drifting attention, and adapts the lesson accordingly. Yet, a teacher who faces dozens of video tiles at once hardly receives this feedback and can no longer tell when a learner has stopped attending.

The problem this thesis addresses is the automatic recognition of student engagement, which determines from the webcam view of a student how engaged that student is with the lesson. This would let a teacher see in real time which students are losing focus and let a platform adjust the pace of a lesson, and aggregated over a session or a cohort it would give an objective account of how well the material holds attention.

1.2 Background and Problems of Research

Vision-based engagement recognition estimates a learner's engagement from a video of that learner. Most work casts it as a classification task, in which a short webcam clip of one student is assigned one of four ordinal engagement levels ranging from highly disengaged to highly engaged. Two main approaches have been pursued. The first encodes each frame with a facial-behaviour toolkit, most often OpenFace [3], which estimates gaze direction, head pose, facial landmarks, and action-unit intensities. The resulting feature sequence is then passed to a classical classifier or a recurrent network, as in an OpenFace+BiLSTM pipeline [4] or in classical models over summary features [5]. The second approach learns directly from pixels with deep video models, for instance an EfficientNetV2 backbone followed by an LSTM [6], or it adapts context-modelling ideas from affect recognition [7]. Progress is measured on a small number of datasets recorded in the wild. Two of these are DAiSEE [2], a widely used public dataset that provides raw video, and CMOSE [1], a more recent and larger dataset that is labelled by experts and ships precomputed OpenFace and I3D features. A systematic review [8] finds severe class imbalance and subjective labels to be common and unsolved problems across the field.

Two limitations of this body of work motivate the thesis. The first is that the facial features are passed through a single encoder. In CMOSE a single OpenFace frame

is a 709-dimensional vector, and this vector bundles measurements of very different physical kinds. It contains a few gaze values, hundreds of eye- and face-landmark coordinates, a set of head-pose parameters, and a set of action-unit strengths. These parts also move at different speeds. The eyes flick quickly and noisily, the head turns slowly and smoothly, and the action units fire in short bursts. Joining all 709 numbers and passing them through one sequence encoder forces a single temporal model onto signals that do not share a timescale, and it obscures the contribution of each behavioural channel.

The second limitation is that studies rely on accuracy as their primary metric and lack a reliable evaluation of generalisation. Almost every study reports accuracy in-domain, training and testing on the same dataset, yet the only deployment that matters uses data recorded with different cameras and conditions. The classes are also heavily imbalanced toward the attentive levels, so a model that simply predicts the majority class can reach a high accuracy while learning nothing that transfers. The metric most papers report can therefore hide the very failure it should expose. Distribution shift makes this worse. Facial features are read from the camera’s view of the face, so a change of camera angle between training and deployment moves the features even when the underlying engagement is unchanged, and this kind of transfer is rarely measured.

These two limitations define the gap this thesis targets. The mixed-feature limitation calls for a representation that preserves the structure of the facial features instead of flattening it. The generalisation limitation calls for an evaluation that measures transfer directly, using metrics that majority prediction cannot inflate.

1.3 Research Objectives and Conceptual Framework

The objective of this thesis is twofold. The first aim is to improve the recognition of engagement on the four-level ordinal scale beyond what standard single-encoder models achieve. The second aim is to give an accurate account of the problem, above all its strong class imbalance and its behaviour when the data changes. To meet these aims, this thesis proposes a solution direction for each of the two problems identified in Section 1.2, one that concerns how the features are modelled and one that concerns how a model is evaluated.

The first direction of work concerns how the features are modelled, and it preserves the structure of the facial descriptor instead of flattening it into a single block. This responds to the limitation that heterogeneous facial channels are forced through one encoder. The OpenFace descriptor is split into five groups that correspond to distinct facial subsystems, namely gaze, eye landmarks, face

landmarks, head pose, and action units, and each group is encoded on its own by an encoder suited to how that subsystem moves in time. A global I3D motion stream can be added alongside the five groups. The resulting embeddings are then joined and classified together, and every stream also carries its own small head so that each channel stays useful on its own. This proposal is the feature-group hybrid model developed in Chapter 3 (Figure 3.4), and it is the main model this thesis puts forward against the single-encoder baselines. Because it is not obvious in advance which encoder best suits each group, or whether the added motion stream helps, this thesis examines these choices across the full design space of the model rather than fixing them by hand.

The second direction of work concerns how a model is evaluated, and it applies an evaluation designed for imbalanced ordinal labels and for deployment. This responds to the limitation that accuracy is used as the primary metric while generalisation is left untested. Every model is selected and ranked on quadratic-weighted kappa (QWK), an ordinal-agreement metric that majority prediction cannot inflate, while balanced accuracy and balanced ordinal error are reported only as secondary checks. Generalisation is then measured directly rather than assumed. Every model is trained and tested across three datasets in a cross-dataset protocol, and one of these datasets was collected for this thesis to present recording conditions that the public data never shows. Together these choices form the evaluation framework under which the proposed model and the baselines are compared.

1.4 Contributions

This thesis makes three contributions.

- (i) This thesis proposes a hybrid model that splits the 709 OpenFace features into five groups. Each group is encoded by its own TCN, Transformer, or LSTM, a global I3D motion stream is optionally added, and every stream is trained with an auxiliary head (Section 3.4). All 243 encoder assignments are tried, with the I3D stream enabled and disabled. The configurations that enable the I3D stream score higher than the strongest baseline, with a median QWK of 0.553 against 0.537 and a best of 0.605. The gain belongs to the whole family rather than to one configuration, and head pose is the only group with a clear encoder preference, namely a TCN, which holds under distribution shift (Section 4.2).
- (ii) A test-only dataset of 366 webcam clips was collected for this thesis. The clips were cut to a fixed length, filtered for tracking quality, labelled by hand, and processed with the same OpenFace and I3D pipeline as the public datasets. This dataset measures performance on real students recorded on their own

webcams, where camera angles and conditions differ from the public datasets (Section 3.2).

- (iii) This thesis runs a 3×3 cross-dataset study, training on CMOSE, DAiSEE, or their combination, and testing on each public test set together with the private set. The study shows that engagement models do not transfer to a new dataset, that accuracy hides this collapse while QWK exposes it, and that combined training is the simplest effective fix. On the private set the architecture and the combined training reinforce each other. The proposed hybrid model trained on both datasets is the best model, at QWK 0.379 against 0.285 for the best baseline, a wider gap than in-domain, which shows the value of the three contributions together (Sections 4.3 and 4.4).

A reproducible pipeline turns about 1500 trained runs into the figures and tables of this thesis. It runs from feature extraction through training and cross-dataset evaluation, and on to the generation of every figure and table.

1.5 Organization of Thesis

The remainder of this thesis is organised as follows.

Chapter 2 reviews vision-based engagement recognition and the background the thesis relies on. It sets the scope and compares the feature-based and end-to-end approaches together with their limits. It then presents the background knowledge that later chapters assume. This background covers the OpenFace and I3D feature extractors, the three temporal architecture families of TCN, LSTM, and Transformer, the loss functions for imbalanced ordinal labels, the ordinal metrics, and the statistical measures used to analyse the results.

Chapter 3 presents the methodology. It gives an overview of the pipeline with diagrams of the baseline and the proposed model. It then describes the three training datasets and their imbalance, the private test set, and the leak-free preprocessing. Finally it defines the single-encoder baselines and builds the feature-group decomposition and the proposed hybrid model on top of them.

Chapter 4 reports the experimental results, centred on the proposed hybrid model. It sets out the loss forms, the training regime, and the 3×3 cross-dataset evaluation matrix. It confirms that the evaluation protocol holds across the model's 486 configurations, and it answers the design questions of which encoder suits which group and whether the I3D stream helps. It then takes the model out of domain, where it outperforms the single-encoder baseline in every cell of the matrix and, with combined training, reaches the best result on the private set.

Chapter 5 draws the findings together, restates the contributions in the light of the evidence, discusses the limits, and proposes future work. The future work includes targeting the rarest disengaged class, applying explicit domain adaptation, and learning the group encoders rather than selecting them by hand.

Appendix A presents the supporting single-encoder baseline study in a protocol-first order. It establishes the metrics of QWK, macro-accuracy, and macro-MAE, the development dataset of CMOSE, and the development loss of cross-entropy. It then reads the in-domain leaderboard and studies prediction agreement to show that the OpenFace and I3D families make different errors. Finally it takes the baselines out of domain in the 3×3 matrix, which sets the reference level against which the proposed model is measured.

CHAPTER 2. LITERATURE REVIEW

2.1 Scope of Research

This thesis is scoped to vision-based engagement recognition, which classifies a learner’s engagement from a frontal webcam recording of one person. Within this scope the thesis concentrates on four parts, and this chapter is organised around them. These parts are the extraction of facial-behaviour and motion features, the modelling of the per-frame descriptors in time, the loss functions suited to imbalanced ordinal labels, and the metrics used to evaluate such models.

Several adjacent problems are left out of scope. Physiological sensors and interaction logs are not used, because they require instrumentation beyond a webcam. Facial-expression and emotion recognition are not treated as an end in themselves, and are referenced only where they inform how engagement features are designed. Finally, the thesis works with OpenFace and I3D features at the descriptor level rather than re-training the extractors: CMOSE ships these features precomputed, while for DAiSEE and the private set they are extracted from the raw video with the same toolkits (Section 3.2); how the extractors are trained internally is reviewed for understanding but not re-implemented.

2.2 Related Work

Research on engagement recognition rests on a small number of datasets recorded in the wild, and two of them are used in this thesis. DAiSEE [2] is the most widely used. It provides short webcam clips annotated on a four-level ordinal engagement scale under uncontrolled conditions, which makes it realistic but noisy. CMOSE [1] is more recent and larger, and its expert annotation gives it cleaner labels, which is why this thesis takes it as the primary dataset and describes both datasets in full in Section 3.2. A systematic review of the field [8] surveys the common pipelines and names two recurring open problems, severe class imbalance and subjective labels.

One family of methods first reduces each frame to a compact descriptor with a toolkit such as OpenFace [3], and then models the resulting sequence in time. Representative pipelines feed OpenFace action units, gaze, and head pose to a recurrent network, as in an OpenFace+BiLSTM model [4], or they apply dimensionality reduction and resampling before a convolutional or fully-connected classifier [9]. Other work aggregates the features over time and compares classical learners such as support vector machines and gradient-boosted trees [5]. The advantage of this family is that the descriptors are compact, cheap to model, and interpretable, since each dimension names a specific behavioural quantity, and the

reported results confirm that OpenFace descriptors carry a real engagement signal. The disadvantage is that the 709-dimensional descriptor is treated as one uniform vector and encoded by a single model. This descriptor mixes behavioural channels whose dynamics differ widely, so one temporal model is forced onto signals that do not share a timescale.

A second family learns directly from pixels. An EfficientNetV2 [10] backbone followed by an LSTM [6] encodes appearance in each frame and integrates it over time, and context-aware affect-recognition models exploit spatial relationships in the wider scene [7]. The advantage of this family is that it captures appearance and motion cues that landmark-based descriptors discard, and it learns its features automatically rather than relying on a fixed toolkit. The disadvantage is that these models are heavier to train, require the raw video rather than precomputed features, and are harder to interpret, because the learned representation is not decomposed into nameable behavioural channels.

Considered together, the two families leave the same two limitations, and these limitations are the motivation for this thesis. The first limitation is that both families encode the facial features as a single block rather than by behavioural subsystem, even though the channels have very different timing, so the structure of the descriptor is discarded. The second limitation is shared with the wider field and concerns evaluation. Studies report accuracy in-domain and rarely test generalisation, yet on imbalanced data a high accuracy can hide a model that predicts the majority class and does not transfer to a new recording setting. This thesis is undertaken to address both limitations, by preserving the structure of the facial features through a feature-group model, and by evaluating every model on an imbalance-resistant ordinal metric across several datasets.

2.3 Background Knowledge

This section presents the technical foundations the rest of the thesis depends on: the feature extractors (Section 2.3.1), the temporal architectures (Section 2.3.2), the loss functions (Section 2.3.3), the evaluation metrics (Section 2.3.4), and the statistical measures for result analysis (Section 2.3.5).

2.3.1 Feature Extraction

a, OpenFace 2.0 facial-behaviour descriptors

OpenFace 2.0 [3] is an open-source facial-behaviour analysis toolkit that processes a video frame by frame and, for each frame, estimates four kinds of measurement: the two- and three-dimensional facial landmarks that outline the face and eye regions, the head pose recovered from those landmarks, the eye-gaze

direction vectors and angles for both eyes, and the intensity and presence of the facial action units of the Facial Action Coding System. Each frame is also tagged with a tracking-confidence and success flag that downstream pipelines use to discard poorly tracked frames.

In CMOSE the per-frame descriptor used here is 709-dimensional. By the meaning of their columns these numbers split into five feature groups, gaze (8 dimensions), eye landmarks (280), face landmarks (340), head pose (46), and action units (35). This decomposition is formalised in Section 3.4.1, where the different temporal behaviour of the groups is used to motivate the proposed architecture in Section 3.4.

b, I3D motion and appearance features

The Inflated 3D ConvNet (I3D) [11] is a video representation obtained by inflating the two-dimensional convolutional filters of an ImageNet-pretrained image network into three dimensions, so they operate over space and time jointly, and then pretraining on the large Kinetics dataset for action recognition. I3D therefore captures appearance and short-range motion that landmark-based descriptors omit. In the data used here, CMOSE ships a single precomputed 1024-dimensional I3D embedding per clip, and for DAiSEE and the private set the same vector is extracted from the raw video. In every case it is one vector pooled over the whole clip, not a per-frame sequence, so the thesis treats I3D as a global clip-level stream of motion and appearance that complements the OpenFace descriptors.

2.3.2 Temporal Architectures

A clip is a sequence of per-frame OpenFace descriptors, so a model must reduce a variable-length sequence to a single engagement prediction. Three architecture families dominate sequence modelling, and all three are used in this thesis, both as single-encoder baselines (Section 3.3) and as the per-group encoders inside the proposed hybrid model (Section 3.4). They differ in how they let one time step influence another.

a, Temporal Convolutional Networks (TCN)

A Temporal Convolutional Network [12] applies one-dimensional convolutions along the time axis, and two design choices make it well suited to behavioural sequences. First, the convolutions are causal: the output at time t depends only on inputs at times $\leq t$, achieved by padding on the left and removing the surplus on the right. Second, successive layers use exponentially increasing dilation, so a layer at depth l skips 2^l steps and a stack of L layers reaches a receptive field that grows as $\mathcal{O}(2^L)$ with few parameters. Each layer is a residual block (two dilated convolutions

with weight normalisation, a rectified-linear nonlinearity, and dropout, summed with a possibly projected copy of the input) that stabilises deep stacks (Figure 2.1). The result is a strong inductive bias for short, local temporal motifs at multiple scales, which matches signals such as a head nod or an action-unit burst.

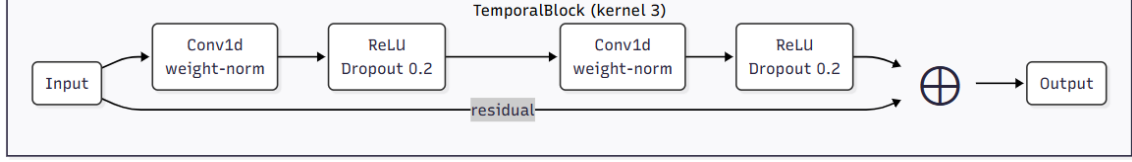


Figure 2.1: Residual TemporalBlock

b, Long Short-Term Memory (LSTM)

The Long Short-Term Memory network [13] is a recurrent architecture that processes the sequence one step at a time while maintaining a cell state c_t and a hidden state h_t (Figure 2.2). Three gates (input i_t , forget f_t , and output o_t) regulate how much new information is written, how much past information is retained, and how much of the cell state is exposed:

$$\begin{aligned} f_t &= \sigma(W_f x_t + U_f h_{t-1} + b_f), & i_t &= \sigma(W_i x_t + U_i h_{t-1} + b_i), \\ o_t &= \sigma(W_o x_t + U_o h_{t-1} + b_o), & \tilde{c}_t &= \tanh(W_c x_t + U_c h_{t-1} + b_c), \\ c_t &= f_t \odot c_{t-1} + i_t \odot \tilde{c}_t, & h_t &= o_t \odot \tanh(c_t), \end{aligned} \quad (2.1)$$

where σ is the logistic sigmoid and $\odot$ is the element-wise product. The gating lets the network carry information across many time steps without the vanishing-gradient problem of plain recurrent networks, giving it a bias toward longer-range dependencies. The cost is that the sequential recurrence is harder to parallelise than convolution or attention.

c, Transformers

The Transformer encoder [14] drops recurrence and convolution and instead relates all time steps directly through self-attention. For queries Q , keys K , and values V obtained by linear projections of the input, attention computes

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V, \quad (2.2)$$

so every output position is a learned weighted combination of every input position, with d_k the key dimension. Because attention is permutation-invariant, order is injected through a positional encoding added to the input; this thesis uses the standard sinusoidal encoding. Multiple attention heads run in parallel and their

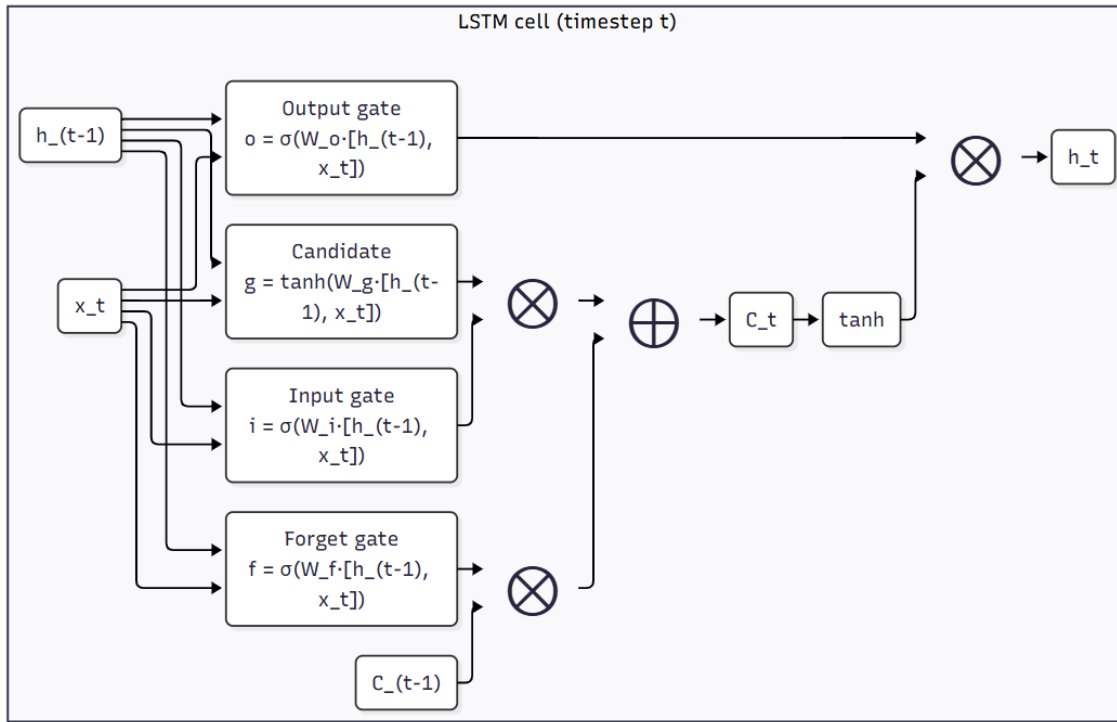


Figure 2.2: LSTM cell at time step t

outputs are concatenated, and each attention sub-layer is followed by a position-wise feed-forward network, with residual connections and layer normalisation throughout (Figure 2.3). The Transformer has the weakest temporal prior of the three families, since it can attend anywhere, which makes it flexible but more reliant on data.

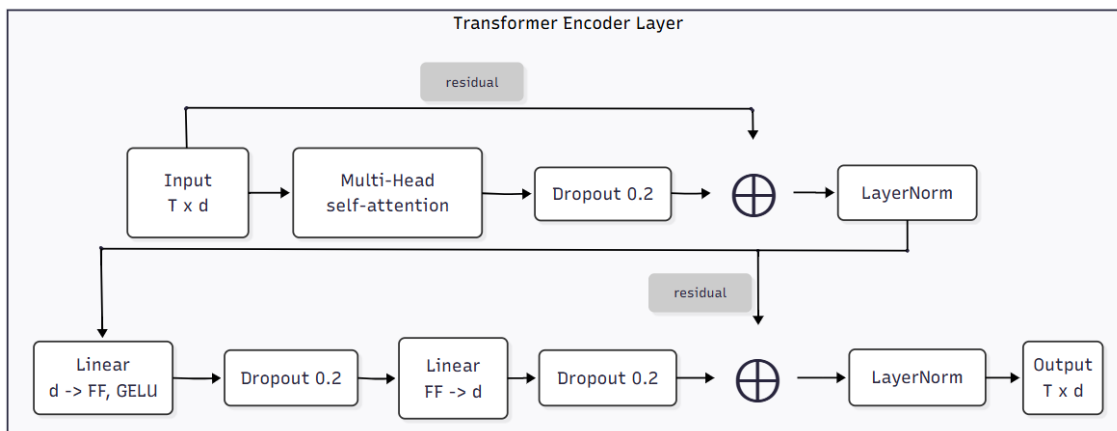


Figure 2.3: Transformer encoder layer

2.3.3 Loss Functions for Imbalance and Ordinality

Let a model produce logits $z \in \mathbb{R}^C$ for the $C = 4$ engagement classes. The softmax function maps these logits to a probability distribution over the classes, the predicted probability of class c being

$$\hat{y}_c = \text{softmax}(z)_c = \frac{e^{z_c}}{\sum_{j=1}^C e^{z_j}} \quad (2.3)$$

so that $\hat{y}_c \in (0, 1)$ and $\sum_{c=1}^C \hat{y}_c = 1$. Let $y \in \{1, \dots, C\}$ be the ordinal ground-truth class, encoded as a one-hot vector through the indicator function $\mathbf{1}[\cdot]$, which returns 1 when its argument is true and 0 otherwise:

$$\mathbf{1}[S] = \begin{cases} 1 & \text{if the statement } S \text{ is true} \\ 0 & \text{if } S \text{ is false} \end{cases} \quad (2.4)$$

so that the one-hot component of the label is $y_c = \mathbf{1}[c = y]$. Three loss families are compared in this thesis; their exact application is given in Section 4.1.2.

Cross-entropy (CE) is the standard classification loss,

$$\mathcal{L}_{\text{CE}} = - \sum_{c=1}^C y_c \log \hat{y}_c, \quad (2.5)$$

which, because y is one-hot, reduces to the negative log-probability of the true class; it treats all classes and all errors as equally severe. On imbalanced data it is dominated by the majority class and on ordinal data it ignores class order, so it serves as the baseline against which the other losses are judged.

Weighted cross-entropy counteracts imbalance by scaling each class's contribution by a weight w_c inversely proportional to its frequency,

$$\mathcal{L}_{\text{WCE}} = - \sum_{c=1}^C w_c y_c \log \hat{y}_c. \quad (2.6)$$

This raises the cost of mistakes on rare classes such as E0 (Highly-Disengage) and stops the model collapsing onto the majority, at the expense of some accuracy. Resampling methods such as SMOTE [15] address the same imbalance at the data level; this thesis instead studies loss-level remedies, which require no change to the data pipeline.

Ordinal (squared-EMD) loss [16] encodes the order of the classes by comparing cumulative distributions. Writing the predicted cumulative distribution as $\sum_{j=1}^k \hat{y}_j$ and that of the one-hot label as the unit step $\sum_{j=1}^k y_j$, the loss is the mean squared distance between the two step functions,

$$\mathcal{L}_{\text{EMD}} = \frac{w_y}{C} \sum_{k=1}^C \left(\sum_{j=1}^k \hat{y}_j - \sum_{j=1}^k y_j \right)^2, \quad (2.7)$$

the squared Earth Mover’s Distance (also called the CDF-MSE) between the predicted and target distributions over the ordered classes [16], not the unsquared 1-Wasserstein distance; the leading w_y is the weight of the true class (Section 4.1.2), set to 1 in the unweighted variant. Because moving probability mass across several classes costs more than moving it to an adjacent class, this loss directly penalises the large ordinal errors ($E0 \leftrightarrow E3$) that accuracy and plain cross-entropy treat as ordinary.

All models are optimised with Adam [17], an adaptive first-order optimiser that maintains per-parameter running estimates of the first and second moments of the gradient.

2.3.4 Evaluation Metrics

Because the labels are ordinal and imbalanced, the choice of metric is itself a research question. This thesis reports six metrics over a test set of N samples with true labels y_i and predicted labels $\hat{y}_i$, but treats exactly one, QWK, as the primary metric that selects the best model; the other five are secondary, reported to describe a result but never to choose one. Where a confusion matrix is needed, O_{ij} denotes the number of samples of true class i predicted as class j .

Accuracy is the fraction of correct predictions,

$$\text{Accuracy} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[y_i = \hat{y}_i]. \quad (2.8)$$

It is reported for comparability with the literature but is not a primary metric here, because on imbalanced data it rewards majority prediction.

Macro-accuracy (balanced accuracy) is the mean of the per-class recalls,

$$\text{MacroAcc} = \frac{1}{C} \sum_{c=1}^C \frac{\sum_{i=1}^N \mathbf{1}[y_i = \hat{y}_i = c]}{\sum_{i=1}^N \mathbf{1}[y_i = c]}, \quad (2.9)$$

giving every class equal weight regardless of frequency. It exposes the minority-class collapse that accuracy hides; here it is a secondary metric, reported to characterise class balance but never used to select a model.

Mean absolute error (MAE) treats the class indices as an ordinal scale and averages the distance between truth and prediction,

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|, \quad (2.10)$$

while macro-MAE averages the per-class MAE so errors on rare classes are not drowned out,

$$\text{MacroMAE} = \frac{1}{C} \sum_{c=1}^C \frac{\sum_{i=1}^N \mathbf{1}[y_i = c] |y_i - \hat{y}_i|}{\sum_{i=1}^N \mathbf{1}[y_i = c]}. \quad (2.11)$$

Both reward predictions that are close on the scale even when not exactly right, and both are smaller-is-better. Macro-MAE is the natural ordinal-distance companion to QWK, with each class weighted equally so the rare disengaged classes are not masked; like the other secondary metrics it is reported alongside QWK and never used to select a model.

Cohen's kappa [18] measures agreement corrected for chance. With observed agreement p_o and expected agreement p_e , $\kappa = (p_o - p_e)/(1 - p_e)$. In its weighted form it is

$$\kappa = 1 - \frac{\sum_{ij} w_{ij} O_{ij}}{\sum_{ij} w_{ij} E_{ij}}, \quad (2.12)$$

where E_{ij} is the expected count under independent marginals and w_{ij} is a penalty matrix. The unweighted variant uses $w_{ij} = 1$ off the diagonal and 0 on it.

Quadratic-weighted kappa (QWK) is the same coefficient with the quadratic penalty

$$w_{ij} = \frac{(i - j)^2}{(C - 1)^2}, \quad (2.13)$$

so disagreements are penalised in proportion to the square of the ordinal distance between the true and predicted classes. QWK suits this problem because it is corrected for chance, respects class order, and cannot be inflated by majority prediction. It is the thesis's primary metric and the only criterion used to select the best model, with macro-accuracy and macro-MAE reported as secondary companions that describe a result but never choose it.

2.3.5 Statistical Measures for Result Analysis

Beyond scoring each model, the results chapters ask three comparative questions that need their own tools: do the six metrics rank the configurations the same way (Section A.2.1), does in-domain strength predict cross-dataset strength (Section 4.3), and do different models make the same per-clip decisions (Section A.4)? The first two are rank-correlation questions; the third is an agreement question.

Spearman’s rank correlation ρ [19] measures how well a monotonic relationship describes two variables by applying the Pearson correlation to their ranks. For n paired observations with rank differences d_i and no ties it reduces to

$$\rho = 1 - \frac{6 \sum_{i=1}^n d_i^2}{n(n^2 - 1)}, \quad (2.14)$$

ranging from -1 (perfectly inverted) through 0 (no monotonic relation) to $+1$ (identical ordering). This thesis uses ρ to test whether a model’s in-domain QWK predicts its QWK on unseen target datasets.

Kendall’s τ [20] measures rank agreement through concordance. Counting the pairs ordered the same way in both variables (concordant, n_c) against those ordered oppositely (discordant, n_d), with no ties,

$$\tau_a = \frac{n_c - n_d}{\binom{n}{2}}, \quad (2.15)$$

again bounded by $[-1, 1]$. It is less sensitive to small samples and to a few large rank displacements than ρ , so it is used here to quantify how far the six metrics agree on the ranking of the baseline configurations.

Prediction agreement asks whether two models agree clip by clip rather than merely score alike, measured by Cohen’s κ (Section 2.3.4) between the two models’ predicted-label vectors; chance-correction matters because two models that lean toward the majority class agree often by default. The headroom from combining a set of M configurations is bounded by the oracle accuracy, the fraction of clips at least one configuration classifies correctly,

$$\text{OracleAcc} = \frac{1}{N} \sum_{n=1}^N \mathbf{1} \left[\exists m \in \{1, \dots, M\} : \hat{y}_n^{(m)} = y_n \right], \quad (2.16)$$

which upper-bounds any fixed combiner and is contrasted with the realistic majority-vote accuracy. A large gap between the best single model, the majority

vote, and the oracle indicates that the models err on different clips and that an ensemble has room to improve.

CHAPTER 3. METHODOLOGY

3.1 Overview

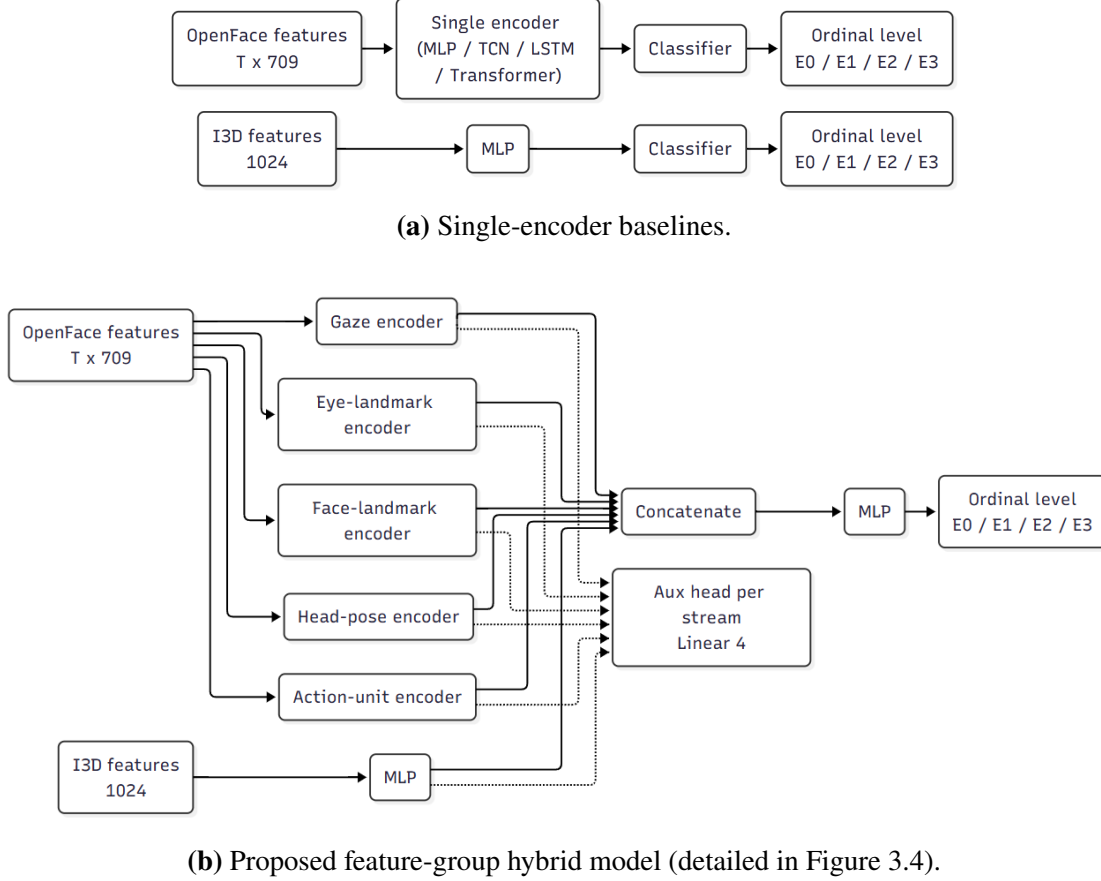


Figure 3.1: The baseline and the proposed hybrid model.

Two public datasets form the training data, while a self-collected set recorded on real webcams serves as an in-the-wild test set (Section 3.2). Every clip carries one of four ordinal engagement levels, from highly disengaged to highly engaged. Each clip is represented by per-frame OpenFace descriptors and a clip-level I3D embedding extracted from the raw video, then resampled to a common length and standardised with training-set statistics so that no target information leaks into preprocessing.

Two model paradigms are compared, contrasted in Figure 3.1. The single-encoder baselines follow the standard approach, in which one encoder, an MLP, TCN, LSTM, or Transformer, consumes the full 709-dimensional OpenFace sequence, or an MLP consumes the pooled I3D vector, before a classifier produces the prediction (Section 3.3).

The proposed hybrid model replaces that single encoder with a split-and-encode design: the OpenFace descriptor is split into five meaning-based feature groups,

each encoded independently by its own temporal encoder, the I3D vector is encoded by an MLP, and the resulting embeddings are concatenated and classified by a shared MLP while an auxiliary head supervises every stream (Section 3.4). The loss functions, training regime, and 3×3 cross-dataset protocol under which every model is trained and evaluated are established with the experiments that use them in Chapter 4.

3.2 Data and Preprocessing

3.2.1 Public datasets

Two public datasets supply the labelled training data. DAiSEE [2] is a widely used engagement dataset of short webcam clips, recorded in the wild, each showing a single learner under uncontrolled conditions and annotated on a four-level ordinal engagement scale, which makes it realistic but noisy; it ships official Train, Validation, and Test partitions used here directly for training, early-stopping validation, and testing. CMOSE [1] is a more recent and larger dataset, likewise collected in the wild but annotated by experts and reporting strong inter-rater agreement, and it ships precomputed OpenFace and I3D features; it is divided into a labelled training split, an `unlabel` split repurposed as the validation set for early stopping and checkpoint selection, and a test split. Both label every clip on the same four ordinal levels used throughout the thesis, E0 (Highly-Disengage), E1 (Disengage), E2 (Engage), and E3 (Highly-Engage), so that CMOSE, with its greater scale and label quality, serves as the primary source and DAiSEE as an independent second source.

Table 3.1: Dataset statistics and class balance (%); the Private set is test-only.

Dataset	Total	Train	Val	Test	E0%	E1%	E2%	E3%
CMOSE	12197	8783	2193	1221	2.800	18.100	69.400	9.600
DAiSEE	8571	5358	1429	1784	0.700	5.100	50.300	43.800
Private	366	0	0	366	2.700	8.500	58.200	30.600

Table 3.1 shows that both public datasets are heavily imbalanced toward the upper-middle engagement levels, and imbalanced in different directions: CMOSE concentrates on the single Engage level, whereas DAiSEE divides more evenly between Engage and Highly-Engage. The disengaged levels are rare in both, yet they are precisely the responses a monitoring system must catch. The same skew holds within each individual split, so it reflects the nature of the task rather than an artefact of how either dataset was partitioned. The self-collected private test set in the final

row, though gathered independently, carries the same engaged-heavy imbalance, so the skew is a property of the engagement task and not of any single collection.

3.2.2 Private dataset

The private set exists to measure deployment behaviour rather than in-dataset behaviour. No model is trained or selected on it, so it is a test-only set, and it is sampled from the webcam recordings a deployed model would actually face, so it probes a condition the public datasets leave untested. The reason this matters is geometric: the OpenFace gaze, head-pose, and landmark features are read from the camera’s view of the face, so the same behaviour recorded from a different camera angle produces different feature values. Figure 3.2 shows private clips that differ along several axes: the camera’s height and distance relative to the student’s eye level, its position relative to the screen, its orientation, whether vertical or horizontal, and the background behind the learner, whether the real room, blurred, or a green screen. CMOSE and DAiSEE were recorded under far more uniform setups and so cannot exercise this variation, which makes a small, locally collected, hand-labelled set drawn from realistic online-class recordings the most direct way to check that a model stays fit once the camera geometry changes.

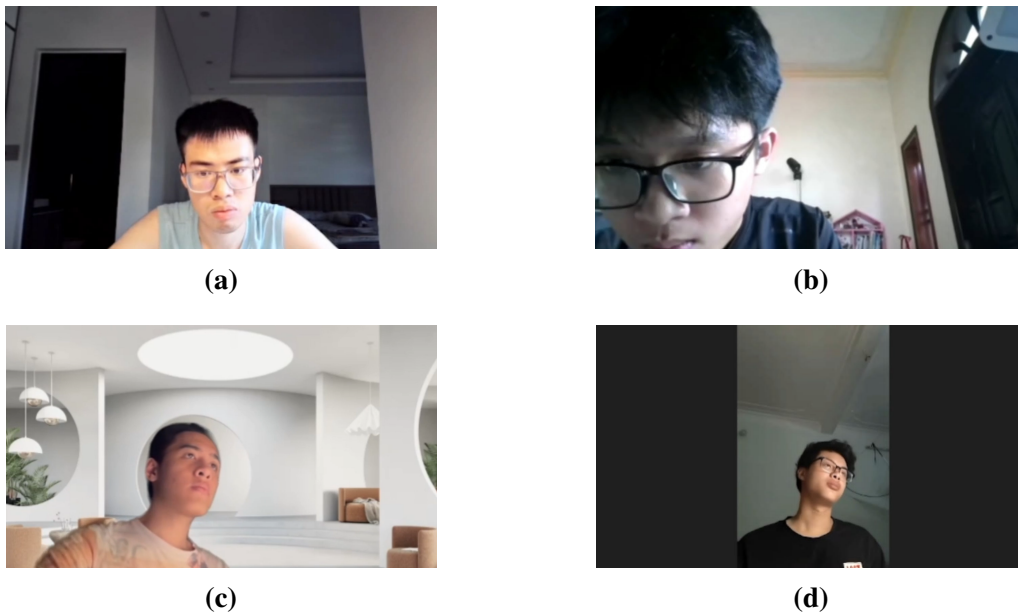


Figure 3.2: Representative frames from the private set, illustrating the range of camera angles found in unconstrained online-class recordings.

The set was built with the same feature pipeline as the public data. Webcam recordings of online-class sessions were sampled at several points and cut into fixed-length ten-second clips, each containing a single learner, and every clip was run through the OpenFace pipeline used for the public datasets, with clips that retained too few tracked frames discarded as unreliable. The accepted clips were

then labelled through a purpose-built local interface, one engagement level per clip and by a single annotator, a reliance revisited as a limitation in Section 5.1. Finally, the same OpenFace (709-d per frame) and I3D (1024-d per clip) extractors used for DAiSEE were applied, so the private features share the format of the precomputed CMOSE features and are distributionally comparable to the training data, and the 366 clips with complete, aligned OpenFace and I3D features form the final set. Its label prior differs again from both public datasets and, together with its independent recording conditions, makes it an in-the-wild measure of deployment behaviour that anchors Section 4.4.

3.2.3 Preprocessing

Each clip is represented by two feature streams. OpenFace 2.0 [3] produces a 709-dimensional descriptor for every frame, and I3D [11] produces a single 1024-dimensional embedding pooled over the whole clip. CMOSE ships both precomputed; for DAiSEE and the private set they are extracted from the raw video with the same toolkits, so all three sources share one feature format. When an OpenFace frame reports more than one detected face the highest-confidence row is kept, and any non-numeric entry is set to zero. Because clips differ in length, each OpenFace sequence is resampled along time to a fixed number of frames by per-feature linear interpolation, 300 frames by default, reduced to 150 for the largest training runs to fit memory and to 75 in the multimodal variant, while the I3D stream, being a single vector per clip, needs no temporal resampling and is encoded by an MLP (Section 3.4).

Every feature is then standardised by a per-feature z -score. The mean and standard deviation are fit on the training split alone, and the validation, test, and private sets are transformed with those training statistics, so no target-set information leaks into preprocessing. The OpenFace and I3D streams are standardised with separate statistics.

3.3 Baseline Models

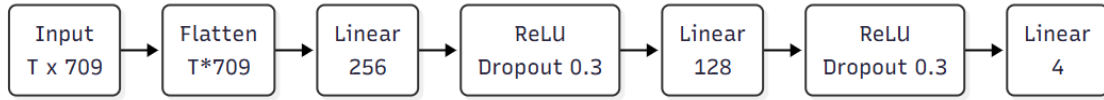
Five single-encoder baselines establish the comparison points, and their architectures are shown in Figure 3.3. Each applies a single encoder to a feature stream and classifies the result.

The multilayer perceptron is the simplest of these encoders, and it serves as the first baseline for both feature streams. An MLP accepts only a fixed-length vector, so the per-frame OpenFace descriptor is flattened into one long vector before the network classifies it (`openface_mlp`), and the I3D clip vector, already fixed in length, is passed to an MLP directly (`i3d_mlp`). Flattening discards the order of the

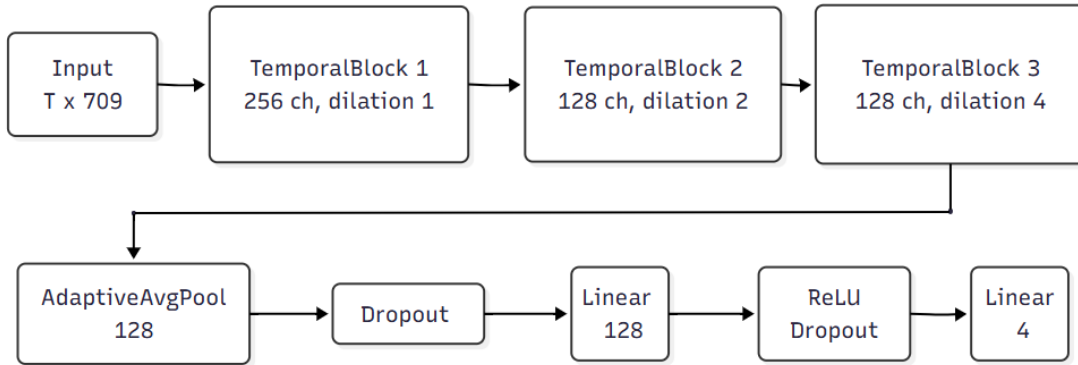
frames, so this baseline reads the clip as an unordered collection of measurements and marks the level that any time-aware model has to improve upon.

The OpenFace descriptor is a sequence, however, and its order carries information that flattening throws away. A clip is an ordered series of frames in the same way that a sentence is an ordered series of words: engagement shows in how the face moves from one frame to the next, not in any single frame read on its own. Three encoders built to read ordered sequences are therefore applied to the raw $T \times 709$ OpenFace stream — a temporal convolutional network (`openface_tcn`), a recurrent network (`openface_lstm`), and a Transformer (`openface_transformer`). Each stands for one of the main families of sequence models, and each preserves the frame order that the MLP removes.

The I3D stream cannot be read in the same way. I3D already averages a whole clip into a single vector when the feature is extracted, so no per-frame sequence remains for a temporal encoder to consume. Its single fixed-length vector fits only an MLP, which is why the I3D stream supports one baseline while the OpenFace stream supports four.



(a) `openface_mlp`.



(b) `openface_tcn`.



(c) `openface_lstm`.

Figure 3.3: The five single-encoder baselines.

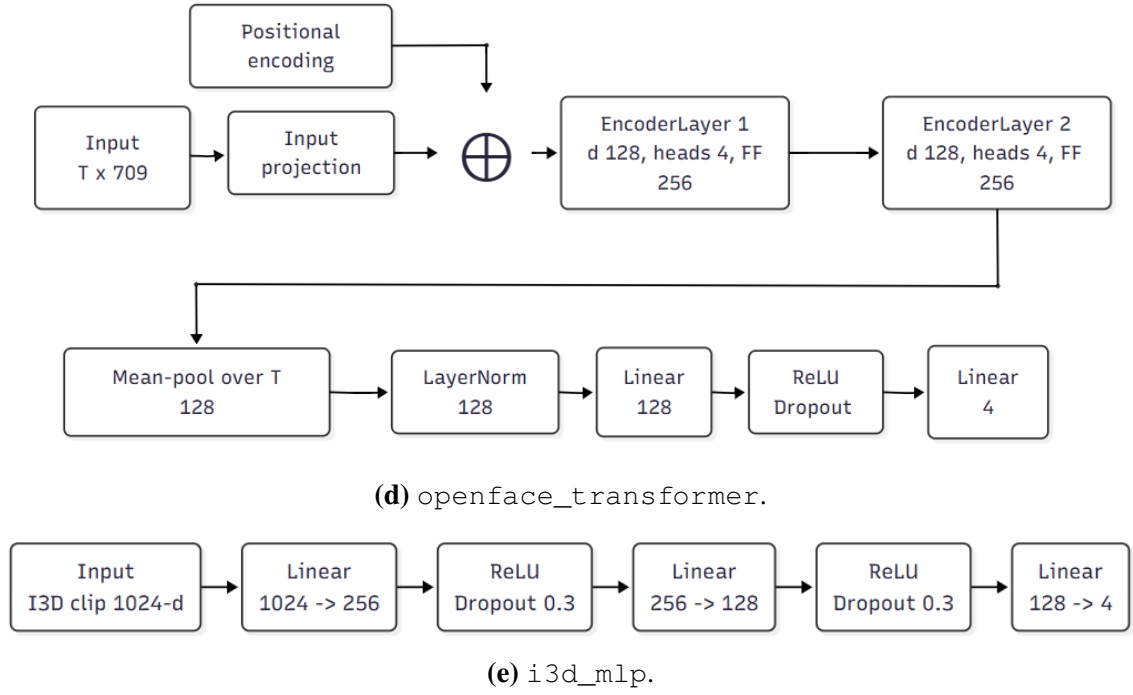


Figure 3.3: The five single-encoder baselines (continued).

3.4 Proposed Hybrid Model

3.4.1 Grouping the facial features by meaning

The four OpenFace baselines share one weakness. A single encoder reads all 709 features at once, and so applies one inductive bias — one built-in assumption about how a signal changes over time — to every feature alike. No single assumption can fit them all, because the 709 features describe several distinct facial subsystems and those subsystems behave in very different ways: gaze and the eye landmarks shift in rapid, jittery micro-movements, head pose drifts slowly and smoothly, and the action units fire in short, sparse bursts. An encoder tuned to track quick eye movements is the wrong instrument for the slow drift of the head, so a single encoder over the whole descriptor suits a few of the features and misfits the rest.

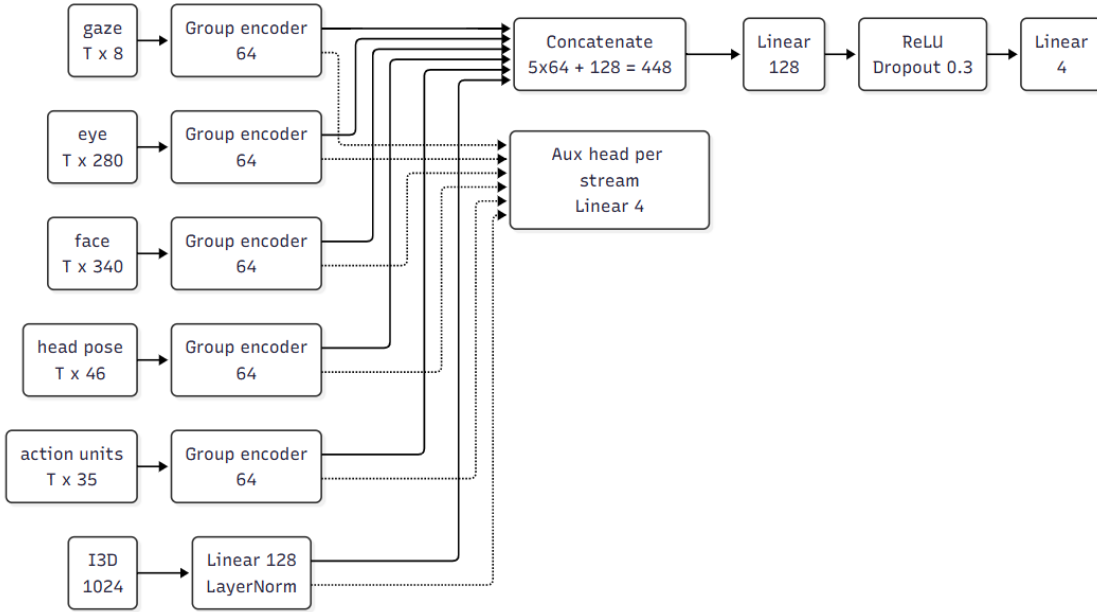
Separating the features by the subsystem they describe removes this conflict. When a group holds only features that behave alike, it can be handed to the encoder whose assumption matches that behaviour, and no encoder is asked to serve incompatible dynamics at once. The 709 OpenFace features are accordingly split into five groups by meaning, matched from the column names in the fixed order used throughout the pipeline: gaze, eye landmarks, face landmarks, head pose, and action units. Table 3.2 lists the grouping, in which every one of the 709 columns belongs to exactly one group. This grouping by meaning is the central idea of the proposed model.

Table 3.2: Grouping of the 709-dimensional OpenFace descriptor into five feature groups by meaning.

Group	Content	Dim
Gaze	gaze direction vectors and angles	8
Eye landmarks	2-D/3-D eye-region landmark coordinates	280
Face landmarks	2-D/3-D facial landmark coordinates	340
Head pose	head translation and rotation parameters	46
Action units	facial action-unit presence and intensity	35
Total		709

3.4.2 Proposed model architecture

The proposed hybrid model is shown in Figure 3.4. It keeps the three encoder families used by the OpenFace baselines — the TCN, the LSTM, and the Transformer — but instead of applying one of them to the whole $T \times 709$ descriptor, it gives each of the five feature groups its own encoder, combines the results, and supervises every stream with its own auxiliary head. Reusing the same three families is a deliberate choice: it ensures that any gain over the baselines can be traced to the grouping alone, and not to a more powerful kind of encoder.

**Figure 3.4:** Proposed hybrid model architecture (I3D stream enabled; the I3D-stream-disabled variant drops the I3D row).

Each group is read by its own `GroupEncoder`, one of the three interchangeable encoders drawn in Figure 3.5. Whichever encoder a group is assigned, its per-frame outputs are averaged over time and layer-normalised, so that every group leaves a

64-dimensional embedding of the same width regardless of how many raw features it began with. Fixing one common width matters because the groups differ greatly in size, from eight features for gaze to 340 for the face landmarks, and a shared width lets each contribute to the fused representation on an equal footing rather than in proportion to its raw dimensionality.

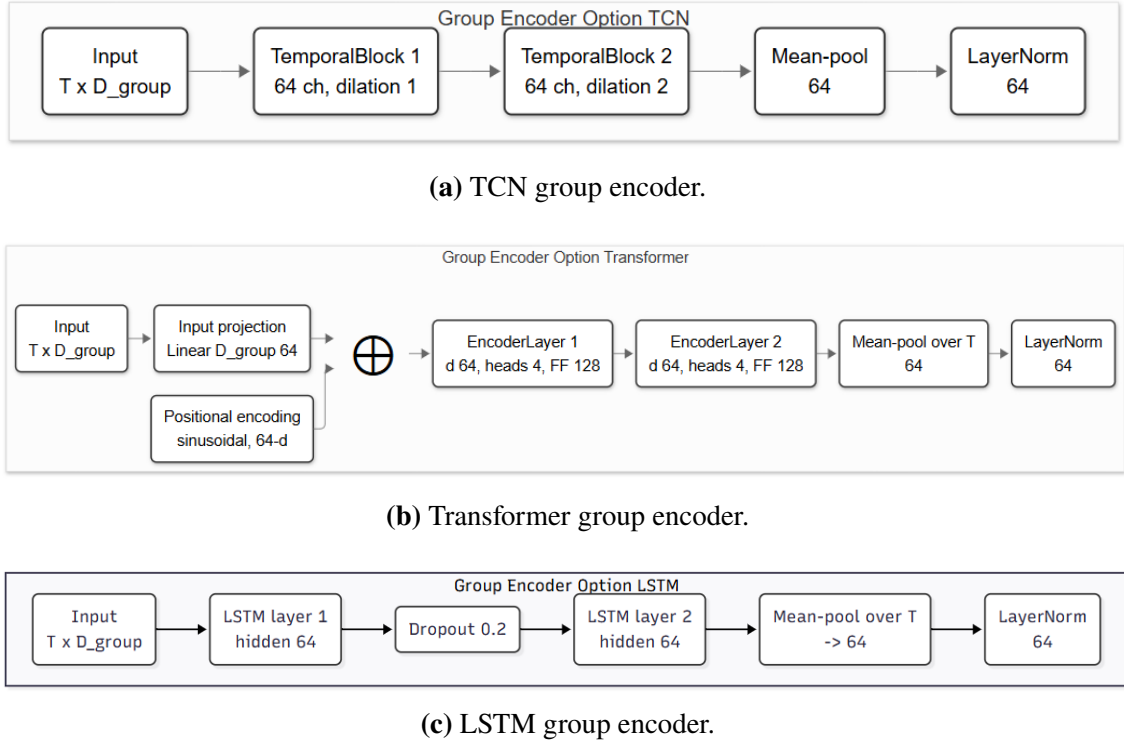


Figure 3.5: The three interchangeable per-group encoders.

The five 64-dimensional embeddings are concatenated into a single 320-dimensional vector, which a two-layer classifier ($320 \rightarrow 128 \rightarrow 4$) maps to the four engagement logits. Each group also feeds a small auxiliary classifier of its own, and the training objective adds a fraction of the averaged auxiliary losses — weighted at 0.2 — to the main loss (Section 4.1.2). These auxiliary heads keep every stream discriminative on its own, so that no single group can be ignored while the others carry the prediction.

An optional sixth stream carries the I3D feature. Because I3D supplies a single pooled vector for the whole clip rather than a per-frame sequence, it is read by an MLP ($1024 \rightarrow 256 \rightarrow 128$) instead of a temporal encoder, and layer-normalised to a 128-dimensional embedding with its own auxiliary head. Enabling this stream widens the fused vector to 448 dimensions ($5 \times 64 + 128$); disabling it leaves the model on the five OpenFace groups alone, and both variants are carried forward for evaluation.

A single configuration of the model is therefore fixed by the encoder chosen for each of the five groups, written as an `arch_key` of five tokens in group order — for example `TCN_T_TCN_LSTM_T` (T = Transformer, TCN = TCN, LSTM = LSTM). With three encoders available to each group this gives $3^5 = 243$ possible configurations, and each can run with the I3D stream enabled or disabled.

CHAPTER 4. EXPERIMENTAL RESULTS

4.1 Experimental Setup

This section states the computing environment, the training hyper-parameters, and the evaluation protocol under which every result in this chapter and in Appendix A is produced. The object of study is the proposed feature-group hybrid model of Section 3.4, evaluated over its full design space and compared against the single-encoder baselines.

4.1.1 Hardware Specification

All models are trained and evaluated on a single desktop workstation, so the whole study runs on commodity hardware rather than a compute cluster. The processor is an Intel Core i5-12400F (six cores, twelve threads, 2.5 GHz base) with 32 GB of system memory, and the accelerator is an NVIDIA GeForce RTX 3060 with 12 GB of dedicated memory. The operating system is Windows 11 Pro. The software stack is Python 3.12 with PyTorch 2.5.1 built against CUDA 12.1 and cuDNN 9.1; training runs on the GPU with automatic mixed precision enabled, so the 12 GB of device memory holds both the single-encoder baselines and the six-stream hybrid model without gradient accumulation. Restricting the study to one workstation keeps the full sweep reproducible on a single machine.

4.1.2 Hyper-parameter Specification

All models are optimised with Adam [17] at a learning rate of 1×10^{-4} . Training uses mini-batches (128 for the baselines, 64 for the hybrid design-space sweep) and early stopping on the validation loss (the `CMOSE_unlabel` split and its analogues) with patience 10 and an epoch cap (800 for the baselines, 400 for the sweep). The checkpoint with the lowest validation loss is restored for testing. Following the convention adopted in this project, only the training and validation loss are recorded per epoch; the six evaluation metrics are computed once, on the held-out test split. Every run uses a fixed seed (42) and mixed precision on the GPU, and every run early-stops well before the cap, so the reported differences are attributable to the architecture rather than to optimisation noise.

Three losses are compared in this thesis, defined in Section 2.3.3. The baseline study fixes cross-entropy as the development loss, because it maximises QWK, the selection metric (Appendix A, Section A.2.3). The proposed model is therefore trained with cross-entropy throughout, so its results isolate the effect of the

architecture from that of the loss, and its per-stream auxiliary heads (Section 3.4) contribute to the objective at weight $\lambda_{\text{aux}} = 0.2$.

A single configuration of the hybrid model is fixed by the encoder assigned to each of the five feature groups, written as an `arch_key` of five tokens in group order, for example `TCN_T_TCN_LSTM_T`. With three encoders available to each group this gives $3^5 = 243$ per-group configurations, each of which runs with the I3D stream enabled or disabled, for 486 configurations in total. The sweep instantiates all of them, so a claim about the architecture is supported by the whole population rather than by one tuned point.

4.1.3 Evaluation Protocol

The proposed model is compared against the single-encoder baselines of Chapter 3, studied in full in Appendix A. Under the protocol of that study the strongest baseline is the OpenFace TCN trained with cross-entropy, at QWK 0.537, the highest of any single encoder on QWK, the selection metric (Appendix A, Section A.3). This value is the reference level the proposed model must exceed, and every comparison in this chapter is measured against it.

Generalisation is measured with a 3×3 matrix: every model is trained on each of {CMOSE, DAiSEE, Combined} and evaluated on each of {CMOSE-test, DAiSEE-test, Private}. The three diagonal cells are the in-domain results; the off-diagonal cells and the Private column measure transfer to unseen distributions. All three training sources are processed through the identical OpenFace and I3D feature pipeline of Chapter 3; the Combined source is the combination of the CMOSE and DAiSEE training splits, and the Private set is the 366 clips collected and hand-labelled for this thesis (Section 3.2.2), used for testing only. This chapter reports the in-domain cell (Section 4.2), the cross-dataset cells (Section 4.3), and the Private column (Section 4.4) in turn.

Each evaluation reports the six metrics of Section 2.3.4: accuracy, macro-accuracy, MAE, macro-MAE, Cohen’s kappa, and QWK. QWK is the single primary metric, and it alone selects models: a reported best model is the argmax of QWK over the tunable axis, ties broken stable-first, with no second metric consulted. The other five are secondary, reported alongside the primary metric to describe how a model treats the rare classes and, in the case of accuracy, to show where it is misleading. The empirical justification for selecting on QWK alone, and for the whole reduced-reporting policy, is given in the baseline study (Appendix A, Section A.2). The figures that survey a full design space or the whole cross-dataset

matrix report all six metrics at once and are captioned as such; the discussion of them stays on the primary metric and its two balanced secondaries.

The study comprises roughly 1500 trained runs: the five baselines under three losses on three sources, and the 486 hybrid configurations on each of the three sources. Every run is trained once with the fixed seed above, so a family-level mean measures robustness to design choices rather than variance across seeds. All runs are aggregated programmatically, and every figure and table in this chapter and in Appendix A is produced from the aggregate by a single command.

4.2 CMOSE In-Domain Result

Trained and tested in-domain on CMOSE, the best configuration of the proposed model exceeds the reference baseline, and the design-space sweep locates where the gain originates. Figure 4.1 reports the full sweep on all six metrics, each panel against the strongest baseline on that metric.

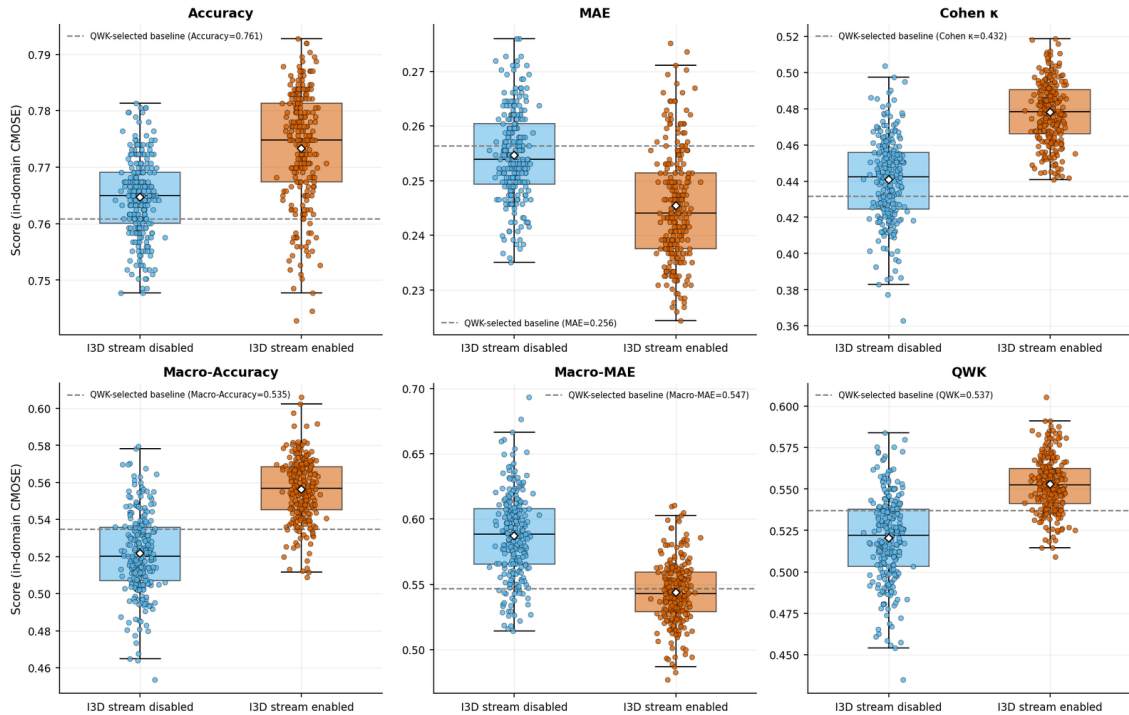


Figure 4.1: Design-space sweep of the proposed hybrid model across all 243 per-group encoder configurations, with the I3D stream enabled and disabled, on all six metrics (in-domain CMOSE). The dashed line is the QWK-selected baseline (OpenFace TCN, cross-entropy).

The six panels confirm that the metric structure of the task carries over to this architecture family. Accuracy is insensitive: every configuration falls within a narrow band around the baseline’s 0.761 (top-left panel), so a model selected on accuracy would see no architecture signal at all. The QWK panel, by contrast, separates the two variants cleanly, which is why QWK is the metric that decides the

comparison and the two balanced secondaries are read alongside it rather than in its place. The same two-block correlation structure found on the baselines (a micro block, a macro block, and QWK the only bridge) reproduces over the 486 in-domain configurations, confirming that it reflects the task and not any one model family.

The design space is almost flat in the encoder assigned to each group. Table 4.1 marginalises each group’s encoder choice, in-domain and pooled over the unseen-target cells. Only head pose shows a clear preference. In-domain it

Table 4.1: Per-group marginal effect of encoder choice on mean QWK, in-domain (CMOSE) and pooled over the unseen-target cells.

Group	Regime	TCN	Transformer	LSTM	Spread	Prefers
Gaze	In-domain	0.539	0.537	0.534	0.006	TCN
Gaze	Unseen targets	0.093	0.095	0.094	0.002	Transformer
Eye landmarks	In-domain	0.538	0.537	0.535	0.003	TCN
Eye landmarks	Unseen targets	0.096	0.092	0.094	0.004	TCN
Face landmarks	In-domain	0.537	0.534	0.539	0.006	LSTM
Face landmarks	Unseen targets	0.092	0.096	0.093	0.004	Transformer
Head pose	In-domain	0.545	0.523	0.541	0.022	TCN
Head pose	Unseen targets	0.100	0.091	0.090	0.010	TCN
Action units	In-domain	0.536	0.537	0.537	0.001	Transformer
Action units	Unseen targets	0.093	0.096	0.093	0.003	Transformer

prefers a TCN (mean QWK 0.545, ahead of LSTM 0.541 and Transformer 0.523, a spread of 0.022), and macro-MAE agrees, lowest under the TCN; the other four groups are insensitive to the encoder (QWK spread ≤ 0.006 , within noise). The preference is consistent across regimes: pooled over the unseen-target cells head pose again has the largest spread (0.010) and again prefers the TCN, while the other groups’ small spreads (0.002–0.004) flip winners arbitrarily. That the head-pose rule survives the shift indicates it reflects signal rather than a tuning accident, since head dynamics such as nodding and turning away are temporally local and suit convolution. Because the rest of the design space is flat, most of the architecture’s benefit must come from decomposing the descriptor and fusing the streams rather than from searching over per-group encoders.

The best configuration exceeds the baseline. Table 4.2 lists the top-five configurations by QWK. The best configuration has the I3D stream enabled, TCN_T_TCN_LSTM_T, and every configuration in the top five mixes at least two encoder types, so the benefit lies in fusing the streams rather than in any single encoder. On QWK, the selection metric, the best proposed-model configuration exceeds the best baseline by +0.068 (0.605 versus 0.537), and the two balanced secondaries agree (macro-accuracy 0.582 versus 0.535, macro-MAE 0.489 versus

Table 4.2: Top-5 configurations of the proposed hybrid model, in-domain (CMOSE), sorted by QWK.

Variant	Arch (gaze_eye_face_head_au)	QWK	Macro-Acc	Macro-MAE	Accuracy
I3D stream enabled	TCN_T_TCN_LSTM_T	0.605	0.582	0.489	0.784
I3D stream enabled	T_TCN_TCN_LSTM_T	0.591	0.606	0.483	0.787
I3D stream enabled	T_T_LSTM_LSTM_TCN	0.591	0.563	0.505	0.781
I3D stream enabled	T_TCN_T_TCN_T	0.588	0.566	0.531	0.789
I3D stream enabled	LSTM_LSTM_TCN_LSTM_T	0.588	0.567	0.517	0.787

0.547), so the improvement is real but moderate. The per-class recalls localise where it arises: relative to the best baseline the proposed model raises recall on the ordinal extreme E3 (Highly-Engage), from 0.38 to 0.51, and on E1 (Disengage), from 0.45 to 0.55, while holding E2 (Engage) near 0.90, and it trades away a little of the rarest class E0 (Highly-Disengage), from 0.40 to 0.37. The gain is therefore uneven: it clears the baseline by strengthening the well-populated upper-engagement region and the middle disengaged class, while the E0 class, at roughly 2–3% of the data, remains the principal open challenge. The next two sections show that this picture becomes more severe out of domain, and that the architecture’s advantage over the baseline widens there.

4.3 Cross-Dataset Result

Across a change of public dataset the ordinal signal does not survive: the proposed model keeps a strong in-domain diagonal but collapses off it, the same shape as the baseline, though it stays ahead of the baseline in every cell. Figure 4.2 shows the proposed model’s 3×3 matrix on all six metrics, each cell the single QWK-selected best configuration profiled across the panels.

The matrix has the same shape as the baseline matrix (Appendix A, Section A.5): a strong diagonal (QWK 0.605 on CMOSE) and an off-diagonal collapse in which QWK and the two balanced secondaries fall together, so the proposed architecture does not rescue cross-dataset transfer. Accuracy remains misleading in exactly the cells where the ordinal signal is gone: on DAiSEE→CMOSE the QWK-selected model reaches only QWK 0.10 yet still scores 0.49 accuracy, because predicting near the majority class scores well on any imbalanced target. A second architecture family thus lands in the same place as the baselines, which confirms that the collapse, and accuracy’s failure to register it, are properties of the task rather than artefacts of the single-encoder models.

Within the collapse, the proposed model is stronger than the baseline in every cell. Figure 4.3 subtracts the baseline matrix (Appendix A, Figure A.5) from the proposed model’s matrix cell by cell, on all six metrics; because both matrices

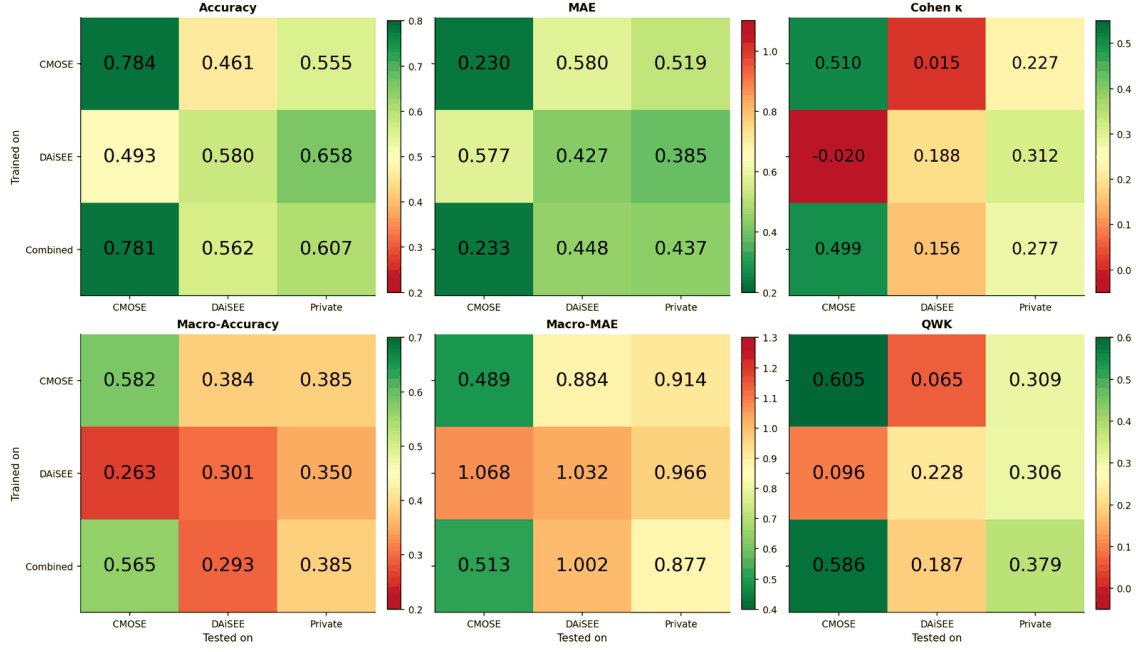


Figure 4.2: Cross-dataset generalisation of the proposed model on all six metrics; each cell is the single QWK-selected best configuration, reported on every metric, so the panels profile the one model deployed in that cell.

select one config per cell by QWK, this is exactly the proposed matrix minus the baseline matrix.

On QWK, the selection metric, every one of the nine train $\times$ test cells is positive: the QWK-selected proposed model outperforms the QWK-selected baseline everywhere. The advantage is smallest on the in-domain diagonal (+0.068 on CMOSE) and largest on the Private column (+0.118, +0.050, +0.094 from CMOSE, DAiSEE, and Combined training). The balanced secondaries agree on the decisive Combined $\rightarrow$ Private cell (macro-accuracy +0.07, macro-MAE -0.12) but scatter in sign on the collapsed public off-diagonal cells, where every metric sits near chance and the direction of a small difference carries little weight. The QWK improvement is therefore a uniform shift of the whole matrix rather than a single tuned point, the strongest form the family-level claim can take, and it shows that distribution shift widens rather than erodes the architecture’s advantage.

That out-of-domain advantage does not come from the I3D appearance stream. Pairing each configuration with its exact I3D-enabled twin (same five per-group encoders, source, and loss) isolates the stream’s effect from the surrounding architecture (Table 4.3). Where the test corpus is seen in training the stream clearly helps (+0.060 mean QWK, 94% of the 972 pairs improving), but across the two public cross-corpus cells its effect turns negative (-0.021 , 26% improving) and on the Private set it is negligible (+0.005, 52%). The I3D embedding is a single

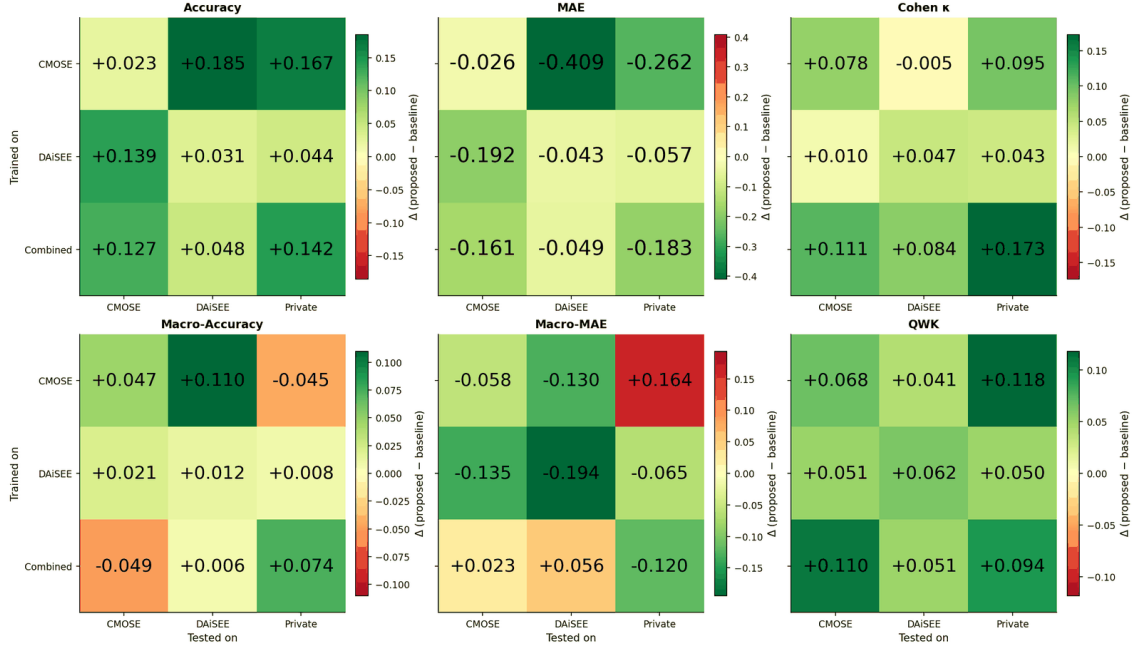


Figure 4.3: Cell-by-cell advantage of the proposed model: QWK-selected best proposed-model minus QWK-selected best baseline over the 3×3 train $\times$ test matrix, on all six metrics (Figure 4.2 minus the baseline matrix, Appendix A, Figure A.5); green is always the proposed-model-favouring direction.

Table 4.3: Paired effect of the I3D stream on QWK, by evaluation regime.

Regime	Pairs (n)	Mean Δ QWK	Pairs improving (%)
Seen target	972	+0.060	94
Cross-corpus (public)	486	-0.021	26
Private (unseen)	729	+0.005	52

clip-level vector of global appearance, which changes with the recording setup, so out of domain it carries dataset bias rather than signal; the OpenFace groups, read in subject-centred units, transfer, and it is they that carry the proposed model’s out-of-domain gain. In practice the I3D stream is worth enabling only when the deployment domain is represented in training.

4.4 Private Dataset Result

The decisive test is the Private set: the 366 clips collected and hand-labelled for this thesis (Section 3.2.2), processed through the identical OpenFace and I3D pipeline and used for testing only. Because no model is trained on it, it is a test-only, in-the-wild set with its own recording conditions and label prior (58% E2, 31% E3, 2.7% E0), so the only available design lever is the training source, and two effects combine there, both already read off the Private column of the cross-dataset matrices. First, the training source matters and the combined dataset transfers best: the best proposed-model QWK on the Private set is 0.379 under combined

training against 0.309 and 0.306 from CMOSE-only and DAiSEE-only training (Figure 4.2, Private column), because the combined source covers a wider range of label priors and recording conditions than either dataset alone. Second, the proposed model beats the best baseline on QWK for every training source and helps more here than in-domain: the advantage is widest under combined training (+0.094, Figure 4.3, Private column), wider than the +0.068 in-domain gain, so distribution shift amplifies the architecture’s advantage. The best Private-set model overall, selected on QWK alone, is the combined-trained proposed hybrid model with the I3D stream enabled, $T_T_T_LSTM_T$, at QWK 0.379.

The per-class F1 scores localise the gain. Figure 4.4 compares the two combined-trained models across the four engagement classes on the Private set.

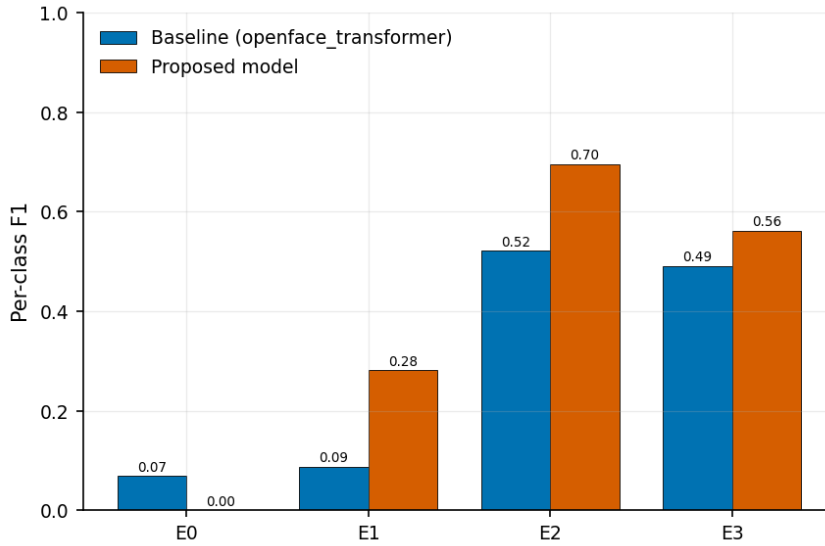


Figure 4.4: Per-class F1 on the Private set for the combined-trained best baseline versus best proposed-model configuration.

The in-domain signature recurs in sharper form: the proposed model lifts F1 on the well-populated E2 ($0.52 \rightarrow 0.70$) and E3 ($0.49 \rightarrow 0.56$) classes and, where the baseline is all but flat, gives the middle E1 class real mass ($0.09 \rightarrow 0.28$), while the rarest E0 class is traded all the way to zero ($0.07 \rightarrow 0.00$, all ten Private E0 clips misread). Out of domain the model concentrates its limited capacity on the engaged region even more sharply, and the rare disengaged extreme remains the principal failure mode in both regimes.

Finally, in-domain strength is a weak predictor of transfer for this family, as it was for the baselines. Pairing each configuration’s in-domain cell (trained and tested on CMOSE) with its deployment cell (trained on Combined, tested on the Private set) gives one point per configuration; Figure 4.5 overlays the 486 proposed-model configurations on the baseline scatter.

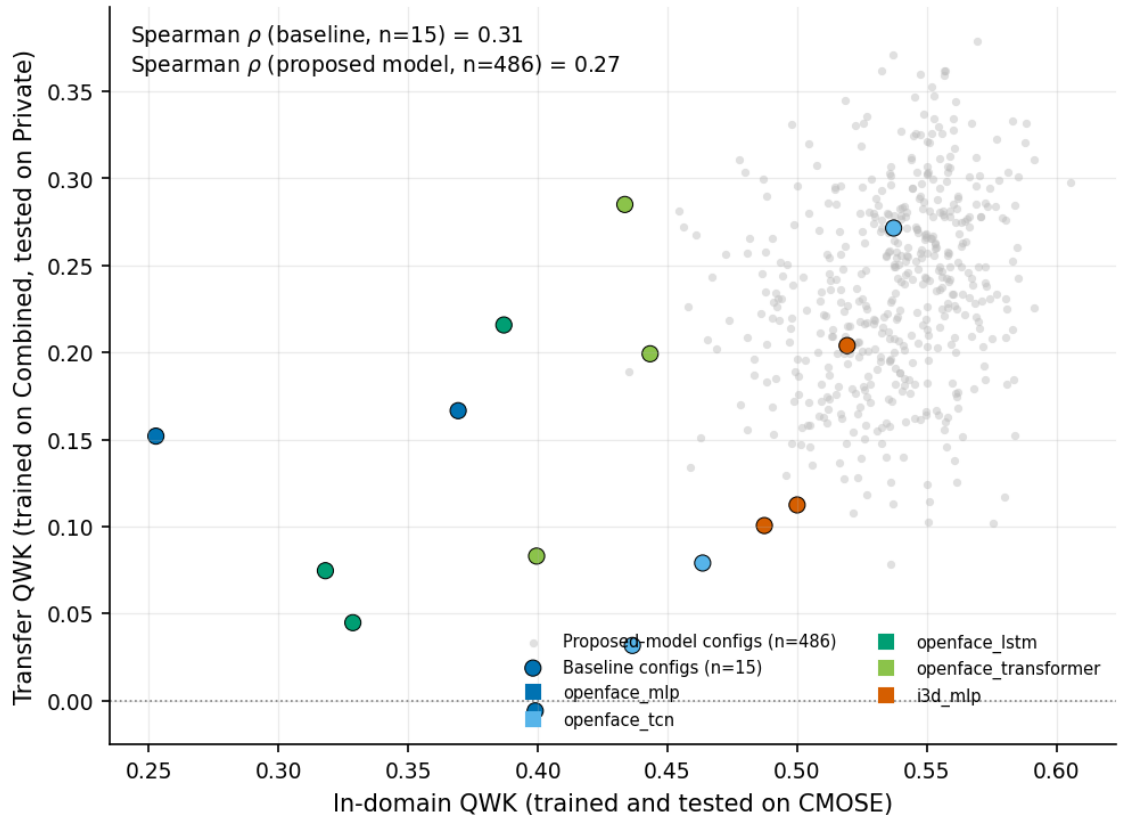


Figure 4.5: In-domain QWK (trained and tested on CMOSE) versus transfer QWK (trained on Combined, tested on the held-out Private set); each configuration is one point pairing its own two runs. Spearman $\rho = 0.27$ for the 486 proposed-model configurations and 0.31 for the 15 baselines.

The correlation stays weak ($\rho = 0.27$, significant only because $n = 486$), in the same range as the baseline relationship ($\rho = 0.31$): a configuration’s in-domain CMOSE ability is at most a weak predictor of how it transfers. The architecture’s contribution shows as position rather than slope. The proposed model’s cloud lies above and to the right of the baseline configurations, reaching both higher in-domain QWK and higher Private-set transfer, so the feature-group design lifts both ends together, while combined training remains the lever that places a model in that upper-right region. A model for an unknown target is therefore chosen by family-level robustness and combined training, not by reading the in-domain leaderboard.

These results follow from the design. Combining datasets improves generalisation by exposing the model to a more diverse set of distributions, which reduces overfitting to either dataset. The proposed model’s margin widens off-domain for the reason it was designed: each behavioural channel is encoded separately and forced to be discriminative through its auxiliary head, so the shared representation depends less on the distinctive features of any one dataset. The two feature families diverge

under shift, as the paired I3D ablation showed (Section 4.3, Table 4.3): fusing the appearance stream adds signal only where the test statistics are represented in training, while the OpenFace behaviour groups carry across setups. The primary result is a clear but moderate improvement: QWK 0.379 exceeds chance, every single-source model, and the best baseline (+0.094), but with a macro-MAE near a full class the model captures the broad engaged-versus-disengaged trend while remaining unreliable at fine distinctions, and the rarest class E0 fails in both regimes. The signal is usable as an aggregate, classroom-level measure but not to flag an individual disengaged learner without human confirmation.

CHAPTER 5. CONCLUSIONS

5.1 Summary

This thesis studied ordinal recognition of student engagement on a four-level scale, using the features OpenFace extracts from each frame and the features I3D extracts from the whole clip. It set out to improve recognition over standard single-encoder models and to describe the problem accurately, in particular its strong, uneven class imbalance and its behaviour under distribution shift. The method is a complete, reproducible pipeline: feature extraction and leak-free preprocessing; the proposed hybrid model, which splits the 709-dimensional OpenFace features into five groups (gaze, eye landmarks, face landmarks, head pose, action units), encodes each group with its own temporal encoder, and optionally adds a global I3D motion stream while training every stream with a small auxiliary head; five single-encoder baselines; three loss functions; and a 3×3 cross-dataset evaluation protocol. The central idea was tested by training all $3^5 = 243$ per-group encoder choices, with the I3D stream enabled and disabled, and by a self-collected, test-only private set of 366 hand-labelled clips that serves as the deciding deployment test. Each of these pieces was chosen so that the two goals, better recognition and an honest account of the problem, could be examined with the same evidence rather than argued separately.

The proposed hybrid model with the I3D stream enabled is the best in-domain model and, as a family, outperforms the strongest baseline. The I3D-stream-enabled configurations are centred above the baseline QWK of 0.537, with a median of 0.553 and 82% of the 243 clearing the baseline, and the best configuration reaches QWK 0.605 with a mixed TCN/Transformer/LSTM assignment. Reporting the family rather than the single best cell is deliberate: because the improvement holds across most of the grid, it reflects the decomposition itself rather than one lucky assignment that might not survive a change of seed or dataset. The gain over the best baseline is nonetheless small, at +0.068 QWK, so the value lies in the framework and the full evidence rather than in a jump in state of the art. Two findings explain why the framework is still worth building. First, the fusion was measured before it was built: the clip-level agreement analysis of Appendix A showed that the best OpenFace model and the I3D baseline disagree with each other more than OpenFace models disagree among themselves, and that each family classifies clips correctly that the other misses, which the paired I3D ablation then confirmed in-domain with a mean gain of +0.060 QWK on the seen-target cells. Second, most groups are not

affected by which encoder they get: only head pose shows a clear preference, for a TCN, with a 0.022 QWK spread across the three encoders against ≤ 0.006 for the other four groups. The design space is therefore flat, a TCN is a safe default for every group, and the gain comes from splitting and fusing the feature groups rather than from searching for a per-group encoder, which makes the design rule far easier to transfer than a single tuned configuration.

Engagement models do not generalise across datasets, and this is the study’s clearest negative result. Off-diagonal QWK drops to near zero, falling to 0.02 from CMOSE to DAiSEE, even when accuracy appears acceptable, so a model that looks competent on its own test set can be worthless on data drawn from a different source. The in-domain leaderboard is of little help in anticipating this, since it predicts transfer only weakly, with a rank correlation of 0.31 for the baselines and 0.27 for the proposed model; a model cannot be selected for deployment from its in-domain score alone. The consequence is a change in how such systems should be measured: ordinal agreement metrics, not accuracy, must be reported on imbalanced engagement data, because accuracy stays high and misleading exactly where it matters least, and training on the combined datasets is the simplest tool that helps when the deployment target is unknown. The self-collected, test-only, in-the-wild private set is where these threads combine into a single deployment result. The proposed model outperforms the best baseline in all nine train $\times$ test cells of the cross-dataset matrix; because the private set is test-only, the only remaining choice is the training data, and training on both datasets together with the proposed hybrid model gives the best result at QWK 0.379, outperforming the best baseline by +0.094 against +0.068 in-domain and surpassing every single-dataset-trained proposed model. That the margin grows from in-domain to the private set is the important point: the same decomposition that is a small in-domain gain becomes a representation that degrades more slowly under distribution shift, which, for a task whose only real test is deployment, is what matters most. Against this stands the one problem the work does not solve. The rare E0 (Highly-Disengage) class, around 2–3% of the data, is traded away rather than solved by the proposed model in-domain and drops to zero recall under distribution shift, so the class that a monitoring system most needs to catch is also the one it detects least reliably.

Several limits should be read alongside these conclusions, and each points to work still to be done. The private set was labelled by a single annotator under the public-dataset rubric, so no inter-annotator agreement can be reported, and its 366 clips limit how reliable the private-set conclusions are, especially for the ten E0 clips, where a single wrong clip strongly affects the recall and the collapse to zero

should be read as directional rather than exact. On the architecture itself, most of the 243-configuration grid is not affected by which encoder it gets and the best in-domain gain over the baseline is small, so the decomposition should be presented as a sound, interpretable framework rather than a large jump in performance. The weak in-domain signal on DAiSEE, at QWK 0.166, meant that dataset could not be used for architecture development because its ordinal signal is too faint to separate architecture differences from label noise, so the cross-dataset conclusions rest mainly on CMOSE and the private set. Finally, a single seed, learning rate, and encoder size were used throughout, so the reported gaps are point estimates rather than confidence intervals, and small differences between nearby configurations should not be over-interpreted.

5.2 Suggestion for Future Work

The most valuable next step is to target the rarest class directly, since it is the one place where the current model clearly falls short. Combining the architecture with fixes aimed at the imbalance, such as ordinal losses tuned for E0, minority oversampling with SMOTE [15], or cost-sensitive decision thresholds, could recover E0 recall without trading away the E3 performance that the in-domain results depend on. The loss study already hints that this is a trade-off rather than a solved problem: cross-entropy gives the best ordinal agreement while weighted and ordinal losses recover minority recall and balanced ordinal error, with no single loss winning on every count. A method that keeps the ordinal-agreement strength of cross-entropy while borrowing the minority-class sensitivity of the weighted and ordinal losses is therefore the natural target. Because E0 is so rare, even small gains here would change the practical value of the system more than further QWK gains on the common classes, and a deployed system would still be wise to treat low-confidence disengagement predictions with care and, where possible, leave the final call to a human.

The cross-dataset gap calls for a dedicated response beyond the plain dataset combination that already helps. Explicit unsupervised domain adaptation or domain-invariant representation learning would attack the transfer collapse head-on rather than diluting it, and a small amount of labelled target data could calibrate the model before deployment where a fully unsupervised approach falls short. Part of the difficulty is that the I3D stream, so useful in-domain, is neutral or slightly harmful under shift because it carries information that is discriminative but specific to each dataset, so a domain-adaptation method that separates the shared engagement signal from dataset-specific motion cues would address both the transfer gap and the fragility of the fusion at once. A related refinement concerns how the group

encoders are chosen: the discrete 243-configuration search could be replaced with a differentiable architecture search or a gating mechanism that picks per-group temporal operators end-to-end, so the model learns which encoder suits which group on its own rather than selecting from a listed grid, and could in principle adapt that choice per domain.

Stronger evaluation would tighten every claim made here and is a prerequisite for turning the study’s directional findings into settled ones. Multi-annotator labelling of a larger private set, with reported inter-annotator agreement, would put the deployment conclusion on firmer ground and give the rare E0 class enough examples for its recall to be estimated with confidence rather than from ten clips. Multi-seed training with confidence intervals would, in turn, replace the point estimates used throughout with statistically grounded claims, making it possible to say which of the small in-domain differences between nearby configurations are real and which fall within noise.

REFERENCE

- [1] C.-H. Wu **and others**, “CMOSE: Comprehensive Multi-Modality Online Student Engagement Dataset with High-Quality Labels,” *in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops* 2024, **pages** 4636–4645. DOI: 10.1109/CVPRW63382.2024.00466
- [2] A. Gupta, A. D’Cunha, K. Awasthi **and** V. Balasubramanian. “DAiSEE: Towards User Engagement Recognition in the Wild,” **urlseen** 26 **june** 2026. **url**: <https://arxiv.org/abs/1609.01885>
- [3] T. Baltrušaitis, A. Zadeh, Y. C. Lim **and** L.-P. Morency, “OpenFace 2.0: Facial Behavior Analysis Toolkit,” *in 13th IEEE International Conference on Automatic Face and Gesture Recognition (FG)* 2018, **pages** 59–66. DOI: 10.1109/FG.2018.00019
- [4] O. Çopur, M. Nakip, S. Scardapane **and** J. Slowack, “Engagement Detection with Multi-Task Training in E-Learning Environments,” *in Image Analysis and Processing – ICIAP 2022* **jourser** Lecture Notes in Computer Science, **volume** 13233, Springer, 2022, **pages** 411–422. DOI: 10.1007/978-3-031-06433-3_35
- [5] S. Malekshahi, J. M. Kheyridoost **and** O. Fatemi. “A General Model for Detecting Learner Engagement: Implementation and Evaluation,” **urlseen** 26 **june** 2026. **url**: <https://arxiv.org/abs/2405.04251>
- [6] F. M. Shiri, E. Ahmadi, M. Rezaee **and** T. Perumal, “Detection of Student Engagement in E-Learning Environments Using EfficientNetV2-L Together with RNN-Based Models,” *Journal on Artificial Intelligence*, **jourvol** 6, **number** 1, **pages** 85–103, 2024. DOI: 10.32604/jai.2024.048911
- [7] L. Zhang, X. Zheng, X. Chen **and** L. Cui, “Three-Dimensional View Relationship-Based Context-Aware Emotion Recognition,” *IEEE Transactions on Neural Networks and Learning Systems*, **jourvol** 36, **number** 7, **pages** 13 567–13 578, 2025. DOI: 10.1109/TNNLS.2024.3476249
- [8] I. Qarbal, N. Sael **and** S. Ouahabi, “Student’s Engagement Detection Based on Computer Vision: A Systematic Literature Review,” *IEEE Access*, **jourvol** 13, **pages** 140 519–140 545, 2025. DOI: 10.1109/ACCESS.2025.3596885
- [9] M. M. Santoni, T. Basaruddin **and** K. Junus, “Convolutional Neural Network Model based Students’ Engagement Detection in Imbalanced DAiSEE Dataset,” *International Journal of Advanced Computer Science*

- and Applications (IJACSA)*, **jourvol** 14, **number** 3, **pages** 617–626, 2023. DOI: 10.14569/IJACSA.2023.0140371
- [10] M. Tan **and** Q. V. Le, “EfficientNetV2: Smaller Models and Faster Training,” *in Proceedings of the 38th International Conference on Machine Learning (ICML)* **jourser** Proceedings of Machine Learning Research, **volume** 139, 2021, **pages** 10 096–10 106.
- [11] J. Carreira **and** A. Zisserman, “Quo Vadis, Action Recognition? A New Model and the Kinetics Dataset,” *in Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)* 2017, **pages** 4724–4733. DOI: 10.1109/CVPR.2017.502
- [12] S. Bai, J. Z. Kolter **and** V. Koltun. “An Empirical Evaluation of Generic Convolutional and Recurrent Networks for Sequence Modeling,” **urlseen** 26 **june** 2026. **url**: <https://arxiv.org/abs/1803.01271>
- [13] S. Hochreiter **and** J. Schmidhuber, “Long Short-Term Memory,” *Neural Computation*, **jourvol** 9, **number** 8, **pages** 1735–1780, 1997. DOI: 10.1162/neco.1997.9.8.1735
- [14] A. Vaswani **and others**, “Attention Is All You Need,” *in Advances in Neural Information Processing Systems (NeurIPS)* 2017, **pages** 5998–6008.
- [15] N. V. Chawla, K. W. Bowyer, L. O. Hall **and** W. P. Kegelmeyer, “SMOTE: Synthetic Minority Over-sampling Technique,” *Journal of Artificial Intelligence Research*, **jourvol** 16, **pages** 321–357, 2002. DOI: 10.1613/jair.953
- [16] L. Hou, C.-P. Yu **and** D. Samaras, “Squared Earth Mover’s Distance-based Loss for Training Deep Neural Networks,” *arXiv preprint arXiv:1611.05916*, 2016.
- [17] D. P. Kingma **and** J. Ba, “Adam: A Method for Stochastic Optimization,” *in International Conference on Learning Representations (ICLR)* 2015.
- [18] J. Cohen, “A Coefficient of Agreement for Nominal Scales,” *Educational and Psychological Measurement*, **jourvol** 20, **number** 1, **pages** 37–46, 1960. DOI: 10.1177/001316446002000104
- [19] C. Spearman, “The Proof and Measurement of Association between Two Things,” *The American Journal of Psychology*, **jourvol** 15, **number** 1, **pages** 72–101, 1904. DOI: 10.2307/1412159
- [20] M. G. Kendall, “A New Measure of Rank Correlation,” *Biometrika*, **jourvol** 30, **number** 1–2, **pages** 81–93, 1938. DOI: 10.1093/biomet/30.1-2.81

APPENDIX

APPENDIX A. BASELINE MODELS

This appendix establishes the two things the main experiment of Chapter 4 builds on. The first is the evaluation protocol used throughout that chapter, the single selection metric, the development dataset, and the training loss, each fixed by a baseline experiment rather than assumed. The second is the evidence that motivates the proposed hybrid model: configurations with near-identical scores disagree on which clips they classify, and the OpenFace and I3D feature families fail on different clips, so learning to fuse them can exceed any single encoder. The baseline rankings themselves enter only as the reference level Chapter 4 must exceed.

A.1 Experimental Setup

Every result in this appendix comes from one apparatus. The five single-encoder baselines of Section 3.3 are each trained under the three losses of Section 4.1.2 on each of the three training sources, optimised identically (Adam, fixed seed, early stopping on validation loss; Section 4.1.2) and evaluated on the held-out test split under the 3×3 cross-dataset protocol of Section 4.1.3. The in-domain experiments (Sections A.2–A.4) use the fifteen CMOSE-trained, CMOSE-tested configurations (five architectures $\times$ three losses); the cross-dataset experiment (Section A.5) uses the full matrix.

Each run is scored on all six metrics of Section 2.3.4, but the reporting is deliberately reduced to follow the protocol. The first experiment (Section A.2.1) settles on QWK as the single metric that selects every best model; from then on the two balanced secondary metrics, macro-accuracy and macro-MAE, accompany each headline result to describe how a model treats the rare classes, while accuracy, micro-MAE, and Cohen’s kappa are reported only where they are themselves the object of study, accuracy where its unreliability is the point (Section A.5) and Cohen’s kappa where prediction agreement is (Section A.4). The overview figures that survey a full design space or the whole cross-dataset matrix (Figure A.3 and Figure A.5 here, Figure 4.1 in Chapter 4) show the complete six-metric grid for reference; their discussion, and every other result, follows this reduced reporting.

Two aggregation conventions hold across this appendix and the experimental results (Chapter 4). A winner is the single configuration with the highest QWK over the tunable axis (here the loss), ties broken stable-first: because QWK is the only selection metric, a reported best model never depends on any other number. A claim about a family of configurations is instead reported as a mean and a distribution over

the design grid. Every configuration is trained once with a fixed seed (Section 4.1.2), so a reported mean measures robustness to design choices, not noise across seeds.

A.2 Evaluation Protocol

Before the baselines can be ranked, three parameters of the protocol must be fixed: the metric that selects models, the dataset the models are developed on, and the loss they are trained with. Each is settled by one experiment below. Two of the arguments need out-of-domain evidence and are stated here as claims, then proved in Section A.5.

A.2.1 Selecting the Primary Metric

The first parameter to fix is the metric that selects models, because on an ordinal, imbalanced task that choice is not neutral. Engagement levels lie on a line, and 69.4% of CMOSE is the single class E2 (Engage) (Table 3.1); accuracy ignores both properties, rewarding majority-class collapse and counting a one-level miss the same as a three-level miss, whereas QWK is chance-corrected and weights every error by the squared distance between predicted and true levels (Section 2.3.4). To test the consequence, the fifteen in-domain baseline configurations are scored on all six metrics, and two questions are put to them: does the leaderboard winner depend on the metric, and do the metrics rank the configurations the same way.

The winner depends on the metric. Table A.1 reports which of the fifteen configurations each metric declares best in-domain, and no two camps agree: QWK picks the OpenFace TCN under cross-entropy (0.537), both macro metrics pick the I3D MLP under the ordinal loss (macro-accuracy 0.635, macro-MAE 0.458), and accuracy, micro-MAE, and Cohen’s kappa all pick the I3D MLP under cross-entropy (0.773, 0.244, 0.436).

Table A.1: Winning baseline configuration by metric (in-domain CMOSE).

Metric	Best model	Loss	Value
QWK	openface_tcn	CE	0.537
Macro-Accuracy	i3d_mlp	Ordinal	0.635
Macro-MAE	i3d_mlp	Ordinal	0.458
Accuracy	i3d_mlp	CE	0.773
MAE	i3d_mlp	CE	0.244
Cohen κ	i3d_mlp	CE	0.436

The metrics also fall into two blocks bridged only by QWK. Figure A.1 reports the Kendall rank correlation between every pair of metrics over the fifteen configurations: a micro block (accuracy and micro-MAE, $\tau = 0.98$, dominated by

the majority class) and a macro block (macro-accuracy and macro-MAE, $\tau = 0.89$) that are almost unrelated to each other (cross-block $\tau \approx 0.01$ – 0.10). QWK is the only metric with moderate-to-strong agreement with both blocks ($\tau = 0.49$ – 0.60).

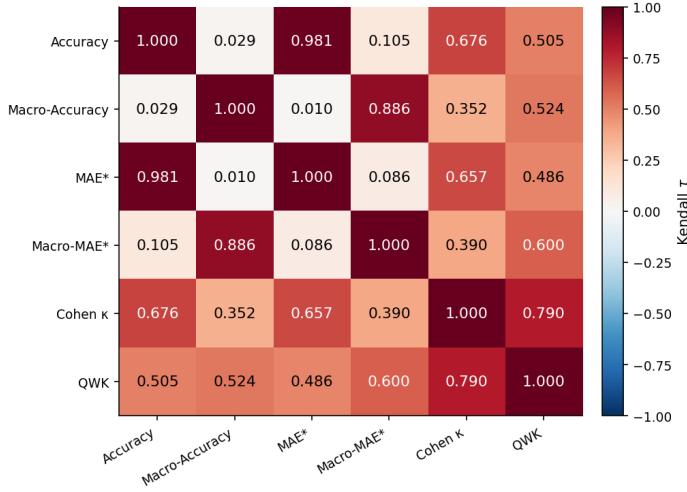


Figure A.1: Rank agreement (Kendall τ) between the six metrics over the fifteen baseline configurations, in-domain on CMOSE (MAE metrics sign-flipped so positive τ means agreement).

The choice of metric therefore changes the result and must be argued, not defaulted. This thesis adopts QWK as its single primary metric, the compromise sensitive to ordinal distance without being dominated by the majority class: the winner is the argmax of QWK, ties broken stable-first, with no second metric consulted. The remaining five are secondary. The decisive argument against accuracy, that it cannot even detect the collapse of a model’s ordinal signal under shift, needs out-of-domain evidence and is deferred to Section A.5. With the metric fixed, the next question is which dataset to develop on.

A.2.2 Selecting the Development Dataset

A development dataset is usable only if models rise meaningfully above chance on it, since a dataset where they do not cannot separate good architectures from bad, and only CMOSE clears that bar. The best in-domain model reaches QWK 0.537 on CMOSE but only 0.166 on DAiSEE, while DAiSEE’s raw accuracy of 0.548 hides the failure, a first sign of how misleading accuracy can be. CMOSE is therefore the development dataset; DAiSEE re-enters as a transfer source and target in Section A.5 and in Chapter 4, where its difficulty is informative rather than obstructive. With the metric and the dataset fixed, the last parameter is the loss.

A.2.3 Selecting the Development Loss

The loss is the axis that trades the primary metric against the balanced secondary ones, so it must be fixed before the architecture is judged. For every baseline, the

loss is swept from cross-entropy through weighted cross-entropy to the ordinal loss, in-domain on CMOSE, and QWK is plotted against the balanced metrics (Figure A.2).

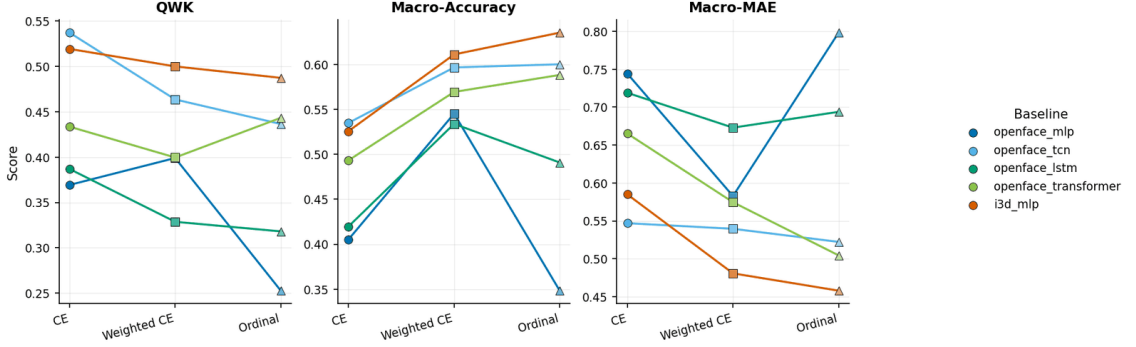


Figure A.2: QWK against the balanced secondary metrics (macro-accuracy and macro-MAE) for every baseline across the three losses, in-domain on CMOSE.

The trade is Pareto: rebalancing the loss raises macro-accuracy and improves macro-MAE while QWK falls for four of the five baselines, so no loss is best on QWK and on the balanced metrics at once. For the I3D MLP, switching from cross-entropy to the ordinal loss lifts macro-accuracy from 0.526 to 0.635 and improves macro-MAE from 0.585 to 0.458 (the best balanced figures of any baseline) while QWK falls from 0.519 to 0.487. Cross-entropy is the development loss because it maximises QWK, the selection metric, and isolates the effect of the architecture from that of the loss (best-architecture QWK by loss: CE 0.537, weighted CE 0.500, ordinal 0.487); the proposed hybrid model of Chapter 4 is consequently trained with it. That this CE > weighted CE > ordinal ranking survives domain shift is the second deferred claim, closed in Section A.5. With the three parameters fixed, the baselines can be ranked.

A.3 In-Domain Results

The aim of this experiment is to set the reference level the proposed architecture must exceed. The five single-encoder baselines are trained and tested in-domain on CMOSE under each of the three losses (four on the OpenFace descriptor: an MLP, a TCN, an LSTM, and a Transformer; and one on the I3D clip vector, the I3D MLP); Figure A.3 reports the complete grid on all six metrics for reference.

On QWK, the selection metric, the strongest baseline is the OpenFace TCN under cross-entropy at QWK 0.537: this is the reference level. The grid adds two structural observations. The loss axis determines on which metric family a model is strongest, cross-entropy on the micro and agreement metrics, the rebalanced losses on the macro metrics that weight the rare disengaged classes equally. The architecture

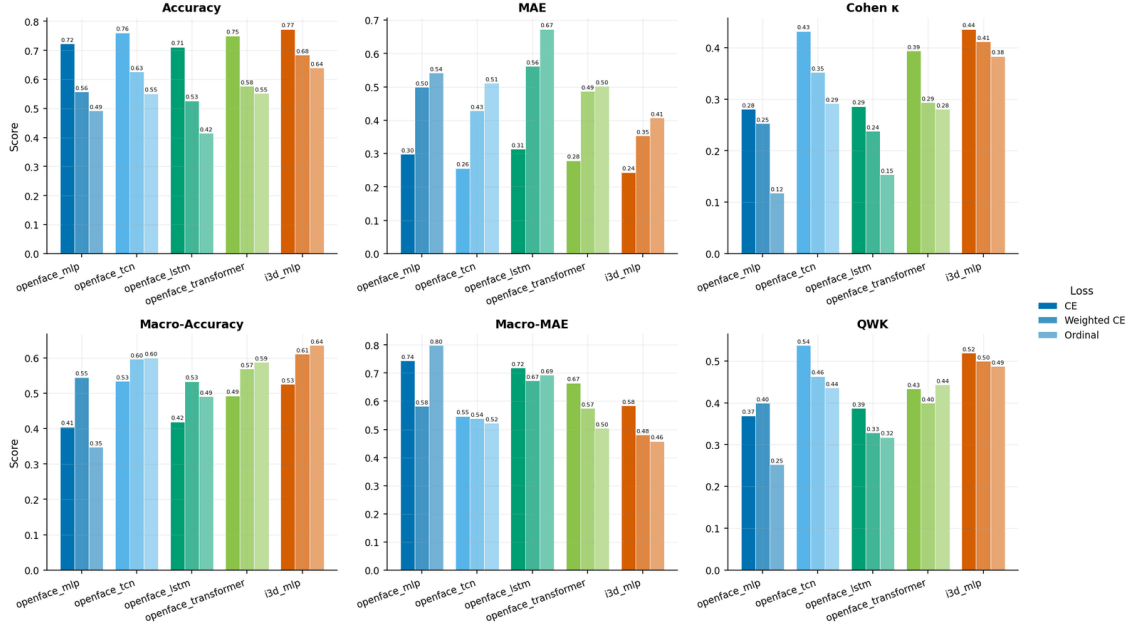


Figure A.3: The five baseline models under the three losses on all six metrics, in-domain on CMOSE.

axis determines which kind of feature is strongest: the OpenFace TCN’s per-frame facial dynamics (a head turning away, gaze drifting off-screen) separate the rarer disengaged levels and rank highest on QWK, while the I3D MLP, over a single pooled appearance vector, aligns with the majority E2 class and ranks highest on every micro metric. On QWK the two score almost the same (0.537 versus 0.519). Two models this close, yet built on disjoint feature families, pose a question the score table cannot answer: are they interchangeable, or do they succeed on different clips?

A.4 Prediction Agreement

Scores summarise how often a model is right, not which clips it is right on. This experiment descends to individual predictions on the 1221 in-domain CMOSE test clips to ask whether models with similar scores make the same decisions, first across the whole grid, then for the two feature families. Cohen’s kappa, the agreement statistic, is the metric of study here.

The fifteen configurations agree far less than their clustered scores suggest. Figure A.4 reports the pairwise Cohen’s kappa between every pair of predictions: mean pairwise agreement is only $\kappa = 0.27$ – 0.39 , and sharing the architecture ($\kappa = 0.391$) or the loss ($\kappa = 0.374$) raises it only mildly over sharing neither ($\kappa = 0.273$). The configurations are different hypotheses about the data, not reparametrisations of one another.

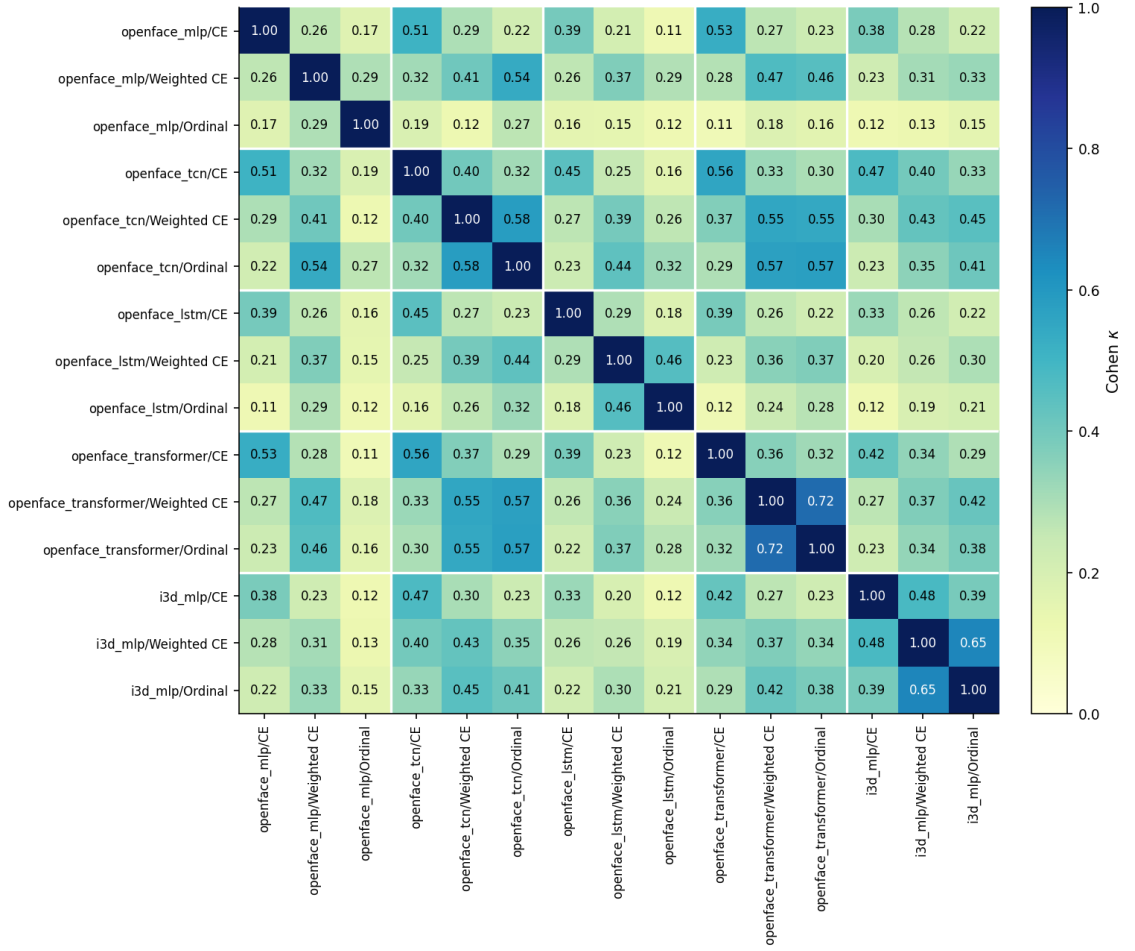


Figure A.4: Pairwise prediction agreement (Cohen’s κ) between all fifteen baseline configurations on the in-domain CMOSE test set, in model-major blocks.

The consequence is headroom. The best single configuration classifies 77.3% of the test clips correctly, but at least one of the fifteen is correct on 97.6% (the oracle), so only 2.4% of clips defeat every model. Plain majority voting over the fifteen lands at 73.5%, below the best single model, because the errors are too decorrelated for unweighted voting to use. Claiming the headroom therefore requires a learned combination of complementary views of the input, which is the design premise of the proposed hybrid model of Chapter 4: it fuses separately encoded behavioural streams instead of averaging finished predictions.

The two feature families scored almost the same, yet differ in nature: OpenFace is a per-frame, structured account of facial behaviour (gaze, eye and face landmarks, head pose, action units), while I3D is a single clip-level vector of global appearance and motion. Whether their close scores mean redundancy or complementarity is the question, and by scores alone the I3D MLP appears merely redundant, since the OpenFace TCN edges it on QWK and on both balanced secondary metrics. The agreement map says otherwise: in Figure A.4 the I3D MLP’s row is the palest in the

grid, its agreement with the OpenFace TCN among the lowest of any pair despite the two having the closest scores. The clip-level decomposition confirms it: against the OpenFace TCN, the I3D MLP has the largest exclusive-correct share of any pair in the grid (8.6% of test clips only I3D gets right) and the smallest both-wrong share (15.3%), and the pair’s oracle accuracy is 84.7%, +7.4 points over the best single model and the largest pair gain in the grid. The behaviour descriptors and the appearance features look at different evidence and fail on different clips, which is the empirical motivation for fusing the two streams in the proposed hybrid model of Chapter 4 that a score table alone could never supply.

A.5 Cross-Dataset Generalisation

In-domain numbers are not the deployment question, and two of the protocol arguments were deferred to out-of-domain evidence. Training each baseline on one dataset and testing on another produces the baseline cross-dataset matrix that Chapter 4 measures the proposed model against, and it closes those two deferred arguments: whether the metric and the loss fixed in-domain remain the right choices once the ordinal signal collapses under a change of dataset.

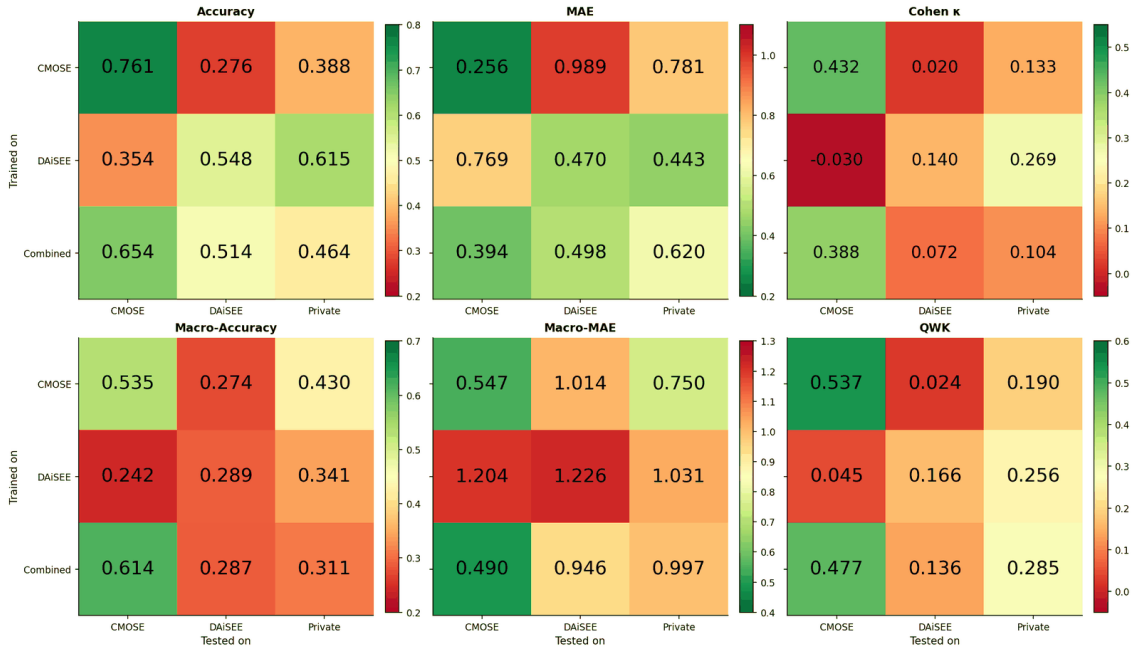


Figure A.5: Cross-domain generalisation of the best baseline per cell, on all six metrics; each cell is the single QWK-selected best baseline, reported on every metric.

The 3×3 heatmaps of Figure A.5 are the baseline that Chapter 4 measures the proposed model against, and they show a cross-dataset collapse. On the diagonal QWK is healthy (CMOSE 0.54), macro-accuracy is above chance (CMOSE 0.54), and ordinal error is moderate (CMOSE macro-MAE 0.55). Off the diagonal all three collapse together: CMOSE→DAiSEE QWK = 0.02 and DAiSEE→CMOSE

$= 0.05$ (chance-level agreement), macro-accuracy falls toward the 0.25 chance floor, and macro-MAE inflates past a full class. The collapse is roughly symmetric, which rules out one dataset being a superset of the other and points to a real shift in both label priors and recording conditions. This is the regime in which the two protocol arguments deferred above can finally be tested.

The cross-dataset matrix supplies the out-of-domain evidence the protocol deferred. Accuracy fails to register the collapse, which completes Section A.2.1. On DAiSEE $\rightarrow$ CMOSE the QWK reads 0.05, chance-level ordinal agreement, yet a model that simply predicted CMOSE’s 69% majority class (Table 3.1) would post 0.69 accuracy, higher than the QWK-selected baseline’s 0.35 in that cell (accuracy panel of Figure A.5). Accuracy would therefore rank a majority-collapsed predictor above the model that actually tracks the ordinal structure, staying comfortable precisely where the ordinal signal is dead. That makes accuracy the least reliable metric here and QWK, the only bridge between the micro and macro blocks, the most.

The loss ranking is preserved under the shift, which completes Section A.2.3. On the unseen-target cells the best-architecture QWK by loss follows the same CE $>$ weighted CE $>$ ordinal order as in-domain. The loss ranking transfers where the architecture ranking does not, so choosing cross-entropy in-domain remains right out of domain, confirming after the fact the choice to establish it as the development loss.